Study of MHD in Accretion Flow Around Black Holes

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Contents

List of Figures

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Abstract

We investigate the global structure of advection-dominated accretion flow solutions (ADAF) around a non-rotating Schwarzschild black hole, where the accretion disk is permeated by large-scale toroidal magnetic fields. The flow is modeled using a magnetohydrodynamic (MHD) framework that incorporates viscosity, mass-loss via winds, and radiative cooling through bremsstrahlung and synchrotron emission. By solving the steady-state conservation equations, we obtain global transonic solutions across a wide range of boundary conditions and parameters, including the accretion rate ($\dot{m}$), viscosity parameter (α_B), and plasma beta (β). We find that, depending on the flow parameters, the accretion solution can exhibit centrifugally supported standing shock waves, leading to the formation of a hot, dense post-shock corona (PSC). Our analysis shows that magnetic fields and radiative cooling significantly influence shock formation. We also discuss how changes in viscous and cooling parameters affect shock stability, and show that magnetic pressure, through reduced plasma β, enhances angular momentum transport and alters the overall disk structure. In addition to determining shock locations, we examine the variation of key flow variables, including density, proton temperature, optical depth, and disk thickness, across the shock transition. Our results indicate that shocks play a vital role in regulating PSC properties. High accretion rates and low plasma β enhance radiative cooling, reduce thermal pressure, and lead to the inward migration of the shock. Additionally, we evaluate the shock characteristics using the compression ratio R, which describes the change in density across the shock, and the shock strength Θ, representing the ratio of pre- to post-shock Mach numbers. These quantities offer insight into the impact of the shock on accretion flow dynamics.

Key words: accretion – accretion discs – black holes – magnetohydrodynamics (MHD) – radiative cooling – shock waves – post-shock corona (PSC).

1 Introduction

Accretion flows are fundamental astrophysical processes in which matter is drawn toward compact objects such as black holes (BHs) and neutron stars under the influence of intense gravitational fields [Shapiro and Teukolsky, 2008]. As matter spirals inward, typically in the form of plasma or gas, it carries angular momentum, leading to the formation of a rotating accretion disk. Accretion is particularly significant because it converts gravitational potential energy into radiation, often producing high-energy emissions such as X-rays observed from compact objects [Yuan and Narayan, 2014]. For black holes, although material disappears beyond the event horizon, it emits considerable radiation during its infall, especially in the inner disk, due to viscous and magnetic interactions [Balbus and Hawley, 1998].

In simplified models, accretion disks are often considered Keplerian, where angular momentum is conserved locally and matter follows nearly circular orbits. However, in realistic astrophysical scenarios, viscous forces are essential for redistributing angular momentum outward, enabling material to gradually spiral inward toward the central object. To account for this, Shakura and Sunyaev [1973] introduced the α-viscosity prescription, where the viscous shear stress is expressed as $W_{r\phi} = \alpha P$. Here, $W_{r\phi}$ represents the $r\phi$-component of the stress tensor responsible for angular momentum transport, P is the total pressure, and α is a dimensionless parameter. While this prescription is widely used to model accretion flows [Matsumoto et al., 1984, Pringle and King, 2007], it does not explain what actually causes the viscosity. The physical origin of viscosity in accretion disks was a long-standing puzzle. A possible explanation was proposed by Balbus and Hawley [1991], who introduced the magnetorotational instability (MRI) as a mechanism capable of driving turbulence in magnetized, differentially rotating flows. This turbulence generates Maxwell stress, which enables efficient angular momentum transport and facilitates the inward drift of accreting matter toward the black hole. Subsequent theoretical and numerical works [Balbus and Hawley, 1998, Hirose et al., 2009, Oda et al., 2007] have reinforced this view, leading to a consensus that magnetic turbulence is primarily responsible for viscous transport in accretion disks, prompting a shift from classical viscosity-based models to a more complete magnetohydrodynamic (MHD) framework.

Compact objects such as BHs and neutron stars are often surrounded by strong magnetic fields, which play a crucial role in shaping the structure and evolution of their accretion disks. These disks, composed of ionized plasma, interact effectively with magnetic fields and are therefore highly responsive to magnetic forces. As a result, magnetic fields influence several important aspects of disk behavior, including angular momentum transport, stability, and energy dissipation [Balbus and Hawley, 1998, Abramowicz and

Fragile, 2013]. Understanding these effects is essential for a complete picture of accretion flow dynamics.

The magnetic field in an accretion disk is typically decomposed into poloidal and toroidal components. While the poloidal component, dominant near the poles, is associated with jet formation [Blandford and Znajek, 1977], the toroidal component, generated and amplified by the disk's differential rotation, dominates the internal disk dynamics [Hirose et al., 2006]. In this study, we focus on a toroidal magnetic field configuration to explore its role in shaping the global structure of the disk. This choice is physically motivated by the shearing motion of the disk, which naturally winds up field lines and enhances the toroidal component.

The toroidal magnetic field impacts accretion disk physics in several critical ways:

- **Angular Momentum Transport:** Weak magnetic fields in differentially rotating disks can destabilize the flow through the magnetorotational instability (MRI), first proposed by Balbus and Hawley [1991, 1998]. Magnetic field lines act like tensioned springs connecting fluid elements at different radii. Differential rotation causes angular momentum to be transferred outward, enabling mass inflow.

- **Pressure Support:** Magnetic fields contribute to the total pressure in the disk. The magnetic pressure is given by $p_m = \frac{B^2}{2\mu_0}$, while the gas pressure is $p_\pi = \frac{R\rho T}{\mu}$, where R is the gas constant, ρ is the density, T is the temperature, and $\mu = 0.5$ is the mean molecular weight. The total pressure is expressed as:

$$P = p_\pi + p_m = p_\pi \left(1 + \frac{1}{\beta}\right), \tag{1}$$

where $\beta = \frac{p_\pi}{p_m}$ is the plasma beta parameter. A high β implies gas pressure dominance, while a low β indicates strong magnetic influence. As matter accretes inward, magnetic field strength increases, β decreases, and magnetic pressure becomes significant, thereby altering the disk's vertical and radial structure [Narayan and Yi, 1995, Yuan and Narayan, 2014, Sarkar et al., 2018, Oda et al., 2010].

- **Thermal Regulation:** Magnetic fields also affect the thermal structure of the disk by governing both heating and cooling processes. The heating is assumed to arise from magnetic energy dissipation, while radiative cooling is dominated by synchrotron and bremsstrahlung emission. In stellar-mass black holes, strong magnetic fields enhance synchrotron cooling [Sarkar et al., 2018], whereas in supermassive black holes, weaker fields make bremsstrahlung cooling more effective [Sarkar and Das, 2016]. These radiative processes regulate the disk luminosity and enable theoretical predictions to be compared with observations. Inverse Compton scattering is excluded from the present study, as it requires a two-temperature accretion model, which lies beyond the scope of this work.

Together, these effects underscore the central role of toroidal magnetic fields in accretion dynamics. They mediate angular momentum transport, support vertical structure, and regulate thermal emission, each contributing to the global behavior of the disk.

In addition to magnetic and thermal processes, radiatively inefficient accretion flows can experience substantial mass loss through winds before matter reaches the black hole. This phenomenon is captured using a mass-loss parameter p, which defines the radial scaling of the accretion rate as $\dot{M}(x) \propto x^p$, where $0 \leq p < 1$. Blandford and Begelman [1999] introduced the Advection-Dominated Inflow-Outflow Solution (ADIOS) as a generalization of the ADAF model, highlighting that only a small fraction of the available gas is ultimately accreted, with the rest being expelled. As p increases, the strength of the outflow grows, significantly modifying the disk's density and structure and reducing the accretion rate near the event horizon.

Given this, we adopt the magnetohydrodynamic (MHD) framework to model accretion disks around black holes. MHD combines fluid dynamics and electromagnetism to describe the evolution of conducting fluids such as astrophysical plasmas. In the following sections, we present a steady-state, axisymmetric model of magnetized accretion flows, incorporating toroidal magnetic fields, viscous transport, radiative cooling, and mass loss due to winds.

In Section 2, we present the fundamentals of magnetohydrodynamics (MHD), including the definition of the magnetic Reynolds number, the distinction between ideal and non-ideal MHD, and the basic cooling mechanisms relevant to accretion disks. Section 3 is dedicated to the derivation of the governing equations, covering mass conservation, momentum balance in all directions, entropy generation, toroidal magnetic flux advection, and the effective optical depth. In Section 4, we explore the formation of shocks in magnetized accretion flows and perform sonic point analysis along with the evaluation of flow variable gradients. Section 5 contains the main results for a stellar mass BH of mass $M_{BH} = 10 M_{\odot}$, where we numerically compute global transonic accretion solutions, both with and without shocks. We analyze how variations in the plasma parameter β and the mass-loss parameter p influence the dynamics of the flow, including shock location, structure, and strength.

2 Magnetohydrodynamics

Magnetohydrodynamics (MHD) is the study of the behavior of electrically conducting fluids like plasmas, where both fluid motion and magnetic fields play a central role. It combines the laws of fluid dynamics and electromagnetism to describe how plasmas evolve under the influence of magnetic and electric fields. MHD is commonly used to understand large-scale astrophysical plasmas such as those found in stars, jets, and accretion disks.

2.0.1 Validity of MHD

Before applying Magnetohydrodynamics (MHD) to accretion disks, it is important to assess its applicability. Fluid dynamics is valid when physical quantities like density ρ, pressure p, and velocity are well-defined over a scale l satisfying $\lambda_{\mathrm{mfp}} \ll l \ll L$, where λ_{mfp} is the mean free path and L a large macroscopic characteristic scale. The dimensionless *Knudsen number* is defined as $K_n = \frac{\lambda_{\mathrm{mfp}}}{L}$ which compares microscopic and macroscopic scales. The condition $K_n \ll 1$ ensures that collisional interactions dominate over free streaming. In our case, using $T = 10^{10}\,\mathrm{K}$, $\rho = 10^{-8}\,\mathrm{g\,cm^{-3}}$, and $L = 10^5 r_g$, we estimate $\lambda_{\mathrm{mfp}} \sim 10^{11}\,\mathrm{cm}$, giving $K_n \sim 0.05$, which is within the acceptable limit for fluid descriptions. [Pringle and King, 2007]

In addition to the Knudsen number, another important criterion is the *Larmor number*, which checks the degree of magnetic coupling. It is defined as the ratio of the Larmor radius r_L to the characteristic macroscopic scale L:

$$\mathcal{L}_r = \frac{r_L}{L}$$

where $r_L = \frac{m v_\perp}{qB}$, with $v_\perp$ the velocity perpendicular to the magnetic field, m the particle mass, q the charge, and B the magnetic field strength. The *Larmor number* tells us how tightly charged particles are confined to magnetic field lines, and when $\mathcal{L}_r \ll 1$, particles are tightly bound to magnetic field lines, allowing the plasma to behave as a magnetized fluid [Goossens, 2003].

For example, for a stellar-mass black hole with $M_{\mathrm{BH}} = 10\,M_\odot$,macroscopic lengthscale $L = 10^5 r_g$, $T = 10^{10}\,\mathrm{K}$, $\rho = 10^{-8}\,\mathrm{g\,cm^{-3}}$, and $B = 10^5$ Gauss, we find $\mathcal{L}_r \sim 10^{-20}$, confirming strong magnetic confinement. Therefore, both K_n and $\mathcal{L}_r$ are sufficiently small to justify the applicability of MHD in this regime.

MHD consists of a system of fundamental equations: the continuity equation, the equation of motion, the induction equation, and the energy equation. Together, they describe the behavior of conducting fluids in magnetic environments.

In most astrophysical systems, the plasma is in motion relative to an observer's (lab) frame. Thus, the relation between electromagnetic fields in the lab frame $(\mathbf{E}, \mathbf{B})$ and the co-moving fluid frame $(\mathbf{E}', \mathbf{B}')$ is important.[1]

Under the assumption that $u \ll c$, the electric field in the fluid frame is approximately:

$$\mathbf{E}' \approx \mathbf{E} + \mathbf{v} \times \mathbf{B}$$

This Galilean form is the foundation for the MHD version of Ohm's law:

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

This equation links current density to electric and magnetic fields in the presence of fluid motion. To track the evolution of the magnetic field, we use Faraday's law:

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E}$$

Using Ohm's law and Ampère's law [2], we arrive at the **MHD induction equation**:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B} \tag{2}$$

Here, $\eta = 1/(\mu_0 \sigma)$ is the magnetic diffusivity. This equation consists of:

- **Advection term**: $\nabla \times (\mathbf{v} \times \mathbf{B})$, representing the transport of the magnetic field by the fluid.
- **Diffusion term**: $\eta \nabla^2 \mathbf{B}$, representing magnetic field dissipation due to finite conductivity.

Together, these terms encapsulate how magnetic fields evolve within a conducting, moving fluid. To better understand the relative significance of these terms, we introduce the Magnetic Reynolds number.

[1] These transformations are derived from special relativity and presented in Appendix A.
[2] neglecting displacement current, justified in Appendix B

2.0.2 Magnetic Reynolds Number

From the induction equation introduced earlier in Eq. (2), we identify two competing processes: the advection of magnetic fields by the fluid flow and their diffusion due to finite conductivity. The relative strength of these terms defines the dimensionless Magnetic Reynolds number R_m, given for a characteristic velocity v_0 and length scale l_0 as:

$$R_m = \frac{\frac{v_0 B}{l_0}}{\frac{\eta B}{l_0^2}} = \frac{v_0 l_0}{\eta}.$$

The Magnetic Reynolds number R_m measures the ratio of magnetic advection to diffusion and plays a crucial role in magnetohydrodynamics (MHD), helping to characterize the evolution of magnetic fields in conducting fluids.

- When $R_m \gg 1$, advection dominates over diffusion, causing magnetic field lines to remain effectively frozen into the fluid and move along with the flow. This regime is particularly relevant for the study of ADAFs.
- When $R_m \ll 1$, magnetic diffusion dominates, allowing magnetic field lines to decouple from the flow and gradually smooth out or dissipate over time.

Understanding the behavior of R_m is essential for analyzing astrophysical systems such as accretion disks, stellar interiors, and solar flares, where the interplay between flow and magnetic fields shapes the dynamics.

2.0.3 Ideal MHD Limit

In the ideal MHD limit, as the conductivity $\sigma \to \infty$, the magnetic diffusivity $\eta \to 0$, eliminating the diffusion term from the induction equation. The Magnetic Reynolds number $R_m \to \infty$, implying that magnetic field advection dominates over diffusion, so the field lines move coherently with the fluid flow. The induction equation reduces to:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B})$$

This leads to the *frozen-in condition*, where magnetic field lines are effectively tied to the motion of the fluid and move with it. This concept is formalized through **Alfvén's**

theorem[3], which states that in a perfectly conducting fluid, magnetic flux through any fluid element is conserved over time.

2.0.4 Non-ideal MHD

When $\eta > 0$, the finite resistivity of the fluid introduces diffusion of the magnetic field and dissipation of magnetic energy. The induction equation takes the full form:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$$

In this regime, magnetic field lines are no longer frozen into the fluid. Instead, they can slip relative to the plasma, enabling processes such as magnetic reconnection, which play a crucial role in phenomena like solar flares, coronal mass ejections, and accretion disk dynamics.

3 Governing Equations

In our work, we examine geometrically thin, axially symmetric accretion discs around a Schwarzschild black hole. Given its axial symmetry and thinness, and assuming that matter accretes through the equatorial plane, we prefer using the cylindrical coordinate system (x, ϕ, z), placing the black hole at the origin. This coordinate system naturally simplifies equations in the thin-disk approximation, allowing separation of vertical and radial dynamics. Due to axial symmetry in cylindrical coordinates, all derivatives with respect to ϕ vanish.

We adopt the geometrical unit system, where $M_{\rm BH} = 2G = c = 1$. Here, $M_{\rm BH}$ is the mass of the black hole under consideration, G is the universal gravitational constant, and c is the speed of light. By this convention, length, time, and velocity can be expressed in units of $r_s = \frac{2GM_{\rm BH}}{c^2}$, r_s/c, and c, respectively, where r_s is the Schwarzschild radius.

In this study, we use the Paczyński-Wiita potential, originally proposed by Paczynsky and Wiita [1980] to approximate the gravitational effects of a Schwarzschild black hole. This potential effectively mimics the behavior of matter near the event horizon without requiring full relativistic equations. It is particularly effective in capturing the innermost disk behavior ($x \lesssim 10$), where general relativistic corrections become important but full

[3] A detailed mathematical derivation that illustrates how magnetic field lines evolve like material line elements and exhibit the frozen-in behavior can be found in Appendix C.

GR treatment is computationally expensive. The Paczyński-Wiita potential at radial distance r is given by:

$$\Phi_{PW}(r) = -\frac{GM}{r - r_s}, \tag{3}$$

Using our unit system, it becomes:

$$\Phi = -\frac{1}{2(x-1)}. \tag{4}$$

where $x = r/r_s$ is the non-dimensional radial distance.

To incorporate the effects of magnetic fields in the accretion flow around the black hole, we model the magnetic field structure in the accretion disc based on global MHD simulations Hirose et al. [2009], Machida et al. [2006], which indicate that the field is turbulent and primarily azimuthal. Following these findings, we represent the magnetic field as a sum of mean and fluctuating components: $B = (0, \langle B_\phi \rangle, 0)$ for the mean field and $\delta B = (\delta B_r, \delta B_\phi, \delta B_z)$ for fluctuations, where $\langle \cdot \rangle$ denotes azimuthal averaging. Since the fluctuating components vanish upon averaging, the azimuthal component dominates over the radial and vertical ones, leading to $\langle B \rangle = \langle B_\phi \rangle \hat{\phi}$ [Oda et al., 2007]. The fluctuating components contribute to angular momentum transport through turbulent viscosity but are neglected in the global structure.

In the accretion disk, fluid flow is described by equations from fluid dynamics. Since we assume a steady state, variables like pressure, density, and velocity remain constant over time, so we focus only on spatial variations. Typically, accretion flow is governed by three hydrodynamic equations i.e. continuity, momentum, and energy conservation. However, the presence of a magnetic field also influences the flow through the radial advection of magnetic flux.

3.1 Continuity Equation (Mass Conservation)

The continuity equation, $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0$, ensures mass conservation. In steady-state and under axial symmetry, it simplifies to:

$$\dot{M} = 2\pi x \Sigma v_x \tag{5}$$

where $\Sigma = \int_{-h}^{h} \rho dz = 2I_n \rho h$ is the vertically integrated density, [Matsumoto et al., 1984] with h being the half-thickness of the disk, ρ is the density at equatorial plane and x is the radial distance from centre of BH. This accretion rate can be expressed in terms of the Eddington rate:

$$\dot{M}_{\text{Edd}} = 1.39 \times 10^{17} \frac{M_{BH}}{M_\odot} \, \text{g s}^{-1}$$

as $\dot{M} = \dot{m} \times \dot{M}_{\text{Edd}}$.

Following Blandford and Begelman [1999], we consider a radial dependence of the accretion rate to account for mass loss (e.g., via outflows):

$$\dot{M} = \dot{M}_{edge} \left(\frac{x}{x_{edge}} \right)^p \tag{6}$$

3.2 Navier-Stokes Equation (Momentum Conservation)

To describe the dynamics of an accreting magnetized fluid, we use the Navier-Stokes equation, which governs momentum conservation. It is derived by applying Newton's second law to a fluid element and accounts for pressure gradients, viscous stresses, gravity, and magnetic fields. In its general form:

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \mathbf{\Pi} + \mathbf{f}, \tag{7}$$

where the total stress tensor $\mathbf{\Pi}^{ij}$ is:

$$\Pi^{ij} = -p_\pi \delta^{ij} + \tau^{ij}, \tag{8}$$

with p_π the gas pressure, τ^{ij} the viscous stress tensor, and δ^{ij} the Kronecker delta. ρ is the mass density and D/Dt the Lagrangian (material) derivative. The force term $\mathbf{f}$ includes both gravitational and magnetic contributions. The gravitational force from the central object (e.g., a black hole) is: $\mathbf{f}_{\text{grav}} = -\rho \nabla \Phi$, and the magnetic force is the Lorentz force:

$$\mathbf{f}_{\text{mag}} = \mathbf{J} \times \mathbf{B}, \quad \text{with} \quad \mathbf{J} = \frac{1}{\mu_0} (\nabla \times \mathbf{B}).$$

Using the Maxwell stress tensor T'_{ij}, the Lorentz force becomes:

$$(f_{\text{mag}})_i = \frac{\partial T'_{ij}}{\partial x_j}, \quad T'_{ij} = \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2} B^2 \delta_{ij} \right), \tag{9}$$

Here, the term $B_i B_j / \mu_0$ represents *magnetic tension*, and $B^2 \delta_{ij} / (2\mu_0)$ represents the *magnetic pressure.* We now combine the gas pressure p_π and the magnetic pressure $p_m = B^2/(2\mu_0)$ into a single scalar total pressure:

$$P = p_\pi + p_m = p_\pi + \frac{B^2}{2\mu_0}. \tag{10}$$

Separately, we define a modified stress tensor containing only the anisotropic (non-scalar) contributions i.e., viscous and magnetic tension terms:

$$M_{ij} = \tau_{ij} + \frac{1}{\mu_0} B_i B_j. \tag{11}$$

Thus, the Navier-Stokes equation becomes:

$$\boxed{\rho \frac{Dv_i}{Dt} = -\partial_i P + \partial_j M_{ij} - \rho\, \partial_i \Phi,} \tag{12}$$

where the scalar pressure gradient $\partial_i P$ captures the isotropic forces from gas and magnetic pressure, while the stress tensor M_{ij} captures the anisotropic effects like shear viscosity and magnetic tension.

3.2.1 Vertical Hydrostatic Balance (Disk Thickness)

To determine the vertical half-thickness $h(x)$ of the accretion disk, we assume that the disk is in vertical hydrostatic equilibrium, which implies that the vertical velocity component $v_z = 0$. Under this condition, the vertical structure of the disk is governed by the vertical component of the Navier-Stokes equation, which reduces to:

$$\frac{1}{\rho} \frac{\partial P}{\partial z} = -\frac{\partial \Phi}{\partial z}, \tag{13}$$

where P is the total pressure, ρ is the mass density, and Φ is the gravitational potential. We adopt the Paczyński-Wiita pseudo-Newtonian potential [Paczynsky and Wiita, 1980], which effectively mimics the general relativistic effects near a Schwarzschild black hole and is given by:

$$\Phi(x, z) = -\frac{GM_{\mathrm{BH}}}{\sqrt{x^2 + z^2} - r_s}, \tag{14}$$

where $r_s = 2GM_{\mathrm{BH}}/c^2$ is the Schwarzschild radius, x is the radial coordinate in the disk plane, and z is the vertical distance from the midplane. Solving this equation yields the scale height $h(x)$, which defines the vertical half-width of the disk i.e., the height above the equatorial plane at which the density becomes negligible.

$$\boxed{h = \sqrt{\frac{2}{\gamma}} a x^{1/2} (x-1).} \tag{15}$$

where a is the adiabatic sound speed defined as:

$$a = \sqrt{\frac{\gamma P}{\rho}}, \tag{16}$$

with P being the pressure, ρ the density, and γ the adiabatic index. We consider $\gamma = 4/3$, applicable for relativistic flow.

The vertically integrated density Σ and pressure W are:

$$\Sigma = \int_{-h}^{h} \rho dz = 2 I_n \rho h \tag{17}$$

$$W = \int_{-h}^{h} P dz = 2 I_{n+1} P h, \tag{18}$$

where $I_n = \frac{(2^n n!)^2}{(2n+1)!}$, h is the half-width, and $n = 1/(\gamma - 1)$ is the polytropic index.[4]

3.2.2 Radial Momentum Equation

To derive the radial component of the momentum equation for a magnetized, axisymmetric accretion flow, we expand the Lagrangian derivative $\frac{D}{Dt} = \frac{\partial}{\partial t} + (\mathbf{v} \cdot \nabla)$ and isolate the x-component. In a geometrically thin disk where the disk half-thickness $h \ll x$, the vertical advection term $v_z\, \partial v_x / \partial z$ becomes negligible due to both the small vertical extent and the weak vertical motion. Assuming steady-state flow, the time derivative $\partial v_x / \partial t$ also vanishes. Under these approximations, the radial component of the momentum equation reduces to:

$$v_x \frac{\partial v_x}{\partial x} - \frac{v_\phi^2}{x} = -\frac{\partial \Phi}{\partial x} - \frac{1}{\rho}\frac{\partial P}{\partial x} + \frac{1}{\rho}(\nabla \cdot \mathbf{M})_x, \tag{19}$$

where $P = p_\pi + p_m$ is the total isotropic pressure, combining gas pressure p_π and magnetic pressure $p_m = B^2/8\pi$. The term $(\nabla \cdot \mathbf{M})_x$ denotes the radial component of the divergence of the anisotropic stress tensor $\mathbf{M}$, which includes viscous stress and magnetic tension. In cylindrical coordinates, this component becomes:

[4] The detailed derivation of the disk thickness is provided in Appendix E.

$$(\nabla \cdot \mathbf{M})_x = \frac{1}{x}\frac{\partial}{\partial x}(xM_{xx}) - \frac{M_{\phi\phi}}{x} + \frac{\partial M_{xz}}{\partial z}. \tag{20}$$

In an axisymmetric configuration dominated by toroidal magnetic fields (B_ϕ), and assuming negligible vertical gradients ($\partial/\partial z \approx 0$), the stress divergence simplifies to:

$$(\nabla \cdot \mathbf{M})_x = -\frac{\langle B_\phi^2 \rangle}{4\pi x}, \tag{21}$$

where angular brackets denote a vertical average. Adopting a pseudo-Newtonian gravitational potential $\Phi = -1/(x-1)$ and defining specific angular momentum $\lambda = xv_\phi$, we obtain the radial momentum equation in the form:

$$\boxed{v_x \frac{dv_x}{dx} = -\frac{1}{\rho}\frac{dP}{dx} + \frac{\lambda^2}{x^3} - \frac{1}{2(x-1)^2} - \frac{\langle B_\phi^2 \rangle}{4\pi \rho x}} \tag{22}$$

This equation highlights the radial force balance between pressure gradients, centrifugal acceleration, gravitational pull, and magnetic tension arising from toroidal field curvature.

3.2.3 Azimuthal Momentum Equation

To derive the azimuthal component of the momentum equation, we isolate the ϕ-component of the Navier-Stokes equation (Eq. 12), yielding:

$$\frac{\partial v_\phi}{\partial t} + v_z \frac{\partial v_\phi}{\partial z} + v_x \frac{\partial v_\phi}{\partial x} + \frac{v_x v_\phi}{x} = \frac{1}{\rho}(\nabla \cdot \mathbf{M})_\phi. \tag{23}$$

The right-hand side involves the azimuthal component of the stress divergence, given by:

$$(\nabla \cdot \mathbf{M})_\phi = \frac{1}{x^2}\frac{\partial}{\partial x}(x^2 M_{\phi x}). \tag{24}$$

Assuming axisymmetry and steady-state flow, the time derivative vanishes. Vertically integrating the equation over the disk thickness and using the specific angular momentum $\lambda = xv_\phi$, we obtain the vertically integrated azimuthal momentum equation:

$$\boxed{\Sigma v_x \frac{\partial \lambda}{\partial x} = \frac{1}{x}\frac{\partial}{\partial x}(x^2 T_{x\phi})} \tag{25}$$

Here, $\Sigma = \int_{-h}^{+h} \rho \, dz$ is the surface density, and $T_{x\phi} = \int_{-h}^{+h} M_{x\phi} \, dz$ is the vertically integrated $x\phi$-component of the stress tensor. In magnetized disks, angular momentum transport is primarily governed by Maxwell stresses, and the dominant stress component becomes:

$$T_{x\phi} = \frac{\langle B_x B_\phi \rangle}{4\pi} h. \tag{26}$$

Following the α-viscosity prescription of Shakura and Sunyaev [1973], this stress can be parameterized as:

$$T_{x\phi} = \frac{\langle B_x B_\phi \rangle}{4\pi} h \tag{27}$$

$$= -\alpha_B (W + \Sigma v^2), \tag{28}$$

where W is the vertically integrated thermal pressure and Σv^2 denotes the vertically integrated ram pressure. This formulation reflects how magnetic tension facilitates the outward transport of angular momentum, enabling inward accretion of mass.

3.3 Entropy Generation Equation

Analyzing entropy generation in accretion flows around black holes provides insight into the thermodynamic mechanisms operating in these extreme environments. As material spirals toward the black hole, irreversible processes such as viscous heating, radiative losses, and angular momentum redistribution contribute to entropy production.

The energy dissipation rate in an accretion disk can be estimated using the viscous stress tensor. The dominant contribution comes from the $r\phi$-component of the stress, $T_{x\phi}$. The viscous heating rate per unit volume is given by the product of this stress component and the strain rate in the disk. For the $x\phi$-plane, the relevant strain rate is $x \, d\Omega/dx$, corresponding to the velocity shear, where $\Omega = v_\phi(x)/x$ is the angular velocity.

This determines the rate of energy dissipation per unit volume and unit time in the disk, expressed as:

$$Q^+ = T_{x\phi}\left(x\frac{\mathrm{d}\Omega}{\mathrm{d}x}\right). \tag{29}$$

The heating of the flow arises from the conversion of magnetic energy into thermal energy through the process of magnetic reconnection [Hirose et al., 2009, Machida et al., 2006], and is therefore expressed by using the stress tensor relation obtained previously in Eq. 28, we get:

$$\begin{aligned} Q^+ &= \frac{\langle B_x B_\phi \rangle}{4\pi} hr\frac{\mathrm{d}\Omega}{\mathrm{d}x} && (30) \\ &= -\alpha_B(W + \Sigma v^2)r\frac{\mathrm{d}\Omega}{\mathrm{d}x} && (31) \end{aligned}$$

To derive the complete form of the entropy generation equation, we apply the first law of thermodynamics per unit mass:

$$\begin{aligned} Td(S/M) &= d(U/M) + p_\pi\, d(V/M) && (32) \\ Tds &= du + p_\pi\, d\left(\frac{1}{\rho}\right) && (33) \\ &= du - \left(\frac{p_\pi}{\rho^2}\right) d\rho, && (34) \end{aligned}$$

Here, $s = S/M$ denotes the *entropy per unit mass*, $u = U/M$ represents the *internal energy per unit mass*, p_π is the *gas pressure*, T is the *temperature*, and ρ is the *mass density*.

Using the relation $u = \frac{p_\pi}{\rho(\gamma-1)}$, we obtain:

$$\boxed{\rho u T\frac{\mathrm{d}s}{\mathrm{d}x} = \frac{u}{\gamma-1}\left[\frac{\mathrm{d}p_\pi}{\mathrm{d}x} - \frac{\gamma p_\pi}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}x}\right] = Q^- - Q^+} \tag{35}$$

Here, Q^+ and Q^- represent the viscous heating and radiative cooling rates per unit volume per unit time, respectively. But before moving forward lets take a look at major cooling process that occur in AF around BH, contributing to cooling rate Q^-.

3.3.1 Cooling in Accretion Disks

Black holes are observed to emit significant electromagnetic radiation, particularly in the X-ray and soft gamma-ray bands. A primary origin of this emission lies in radiative cooling processes that operate within the surrounding accretion flow. As matter spirals in toward the black hole, gravitational potential energy is converted into thermal energy through viscous dissipation and magnetic turbulence. To maintain the inward flow of matter, the thermal energy generated within the accretion disk must be effectively radiated away, making cooling processes essential to the dynamics of accretion. The overall radiation emitted by the disk, characterized by its luminosity $L_{\rm disc}$, is computed by integrating the radial distribution of the cooling rate Q^{-} across the disk:

$$L_{\rm disc} = 4\pi \int_{x_i}^{x_f} Q^{-}(x)\, x\, dx,$$

where x_i and x_f mark the inner and outer boundary of the disk.

In magnetized accretion flows, the dominant radiative cooling processes are:

1. **Bremsstrahlung (free-free) emission**: important in hot, ionized plasma where Coulomb collisions between electrons and ions result in the emission of soft photons.
2. **Synchrotron radiation**: crucial in the presence of strong magnetic fields, especially when relativistic electrons spiral around toroidal magnetic field lines.
3. **Inverse Compton scattering**: significant in the post-shock region or corona, where energetic electrons scatter soft photons to higher energies.

These processes together shape the radiative output and thermal structure of the disk. In optically thin regions, such as the inner parts of advection-dominated flows or post-shock zones, radiative losses occur locally. In contrast, in optically thick regions, photons are trapped and energy is advected inward or diffused out on longer timescales.

Bremsstrahlung Cooling Bremsstrahlung cooling becomes dominant in regions of high density and moderate temperature, where free electrons interact with ions, emitting photons due to acceleration in the Coulomb field. We adopt the emissivity from Shapiro & Teukolsky (1983)Shapiro and Teukolsky [2008]:

$$\Lambda_{ff} = 1.43 \times 10^{-27} \left(\frac{\rho}{m_p}\right)^2 T_e^{1/2} \quad {\rm erg\,cm^{-3}\,s^{-1}}, \tag{36}$$

where T_e is the electron temperature and ρ is the rest-mass density.

In our analysis, we relate T_e to the proton temperature T_p using a temperature scaling relation from [Chattopadhyay and Chakrabarti, 2002], and express the full vertically integrated, dimensionless form of the cooling rate.[5]

$$Q_{Br}^{-} = \frac{1.98 \times 10^{-10}}{m_p^2} \frac{\rho h \dot{m}}{4\pi I_n x^{3/2}(x-1)v} \left(\frac{m_p}{m_e}\right)^{1/4} \left(\frac{\mu m_p}{2k_b}\right)^{1/2} \left(\frac{\beta}{\beta+1}\right)^{1/2} \frac{1}{2GM_\odot c} \tag{37}$$

This expression can be further simplified by separating the physical variables and absorbing the constants into a coefficient, along with a cooling efficiency factor ξ [Sarkar and Das, 2016], yielding:

$$\boxed{Q_{Br}^{-} = \xi \times \frac{C\rho h}{x^{3/2}(x-1)v} \left(\frac{\beta}{\beta+1}\right)^{1/2}} \tag{38}$$

where the C is defined as:

$$C = 1.98 \times 10^{-10} \frac{\dot{m}}{4\pi I_n m_p^2} \left(\frac{m_p}{m_e}\right)^{1/4} \left(\frac{\mu m_p}{2k_b}\right)^{1/2} \frac{1}{2GM_\odot c} \tag{39}$$

Here, ρ is the mass density, h is the half-thickness of the disc, $\dot{m}$ is the accretion rate in Eddington units, v is the radial velocity, β is the plasma beta parameter, I_n is a numerical factor from vertical integration, and ξ is the phenomenological cooling efficiency factor introduced to scale the cooling strength.

Synchrotron Cooling Synchrotron radiation becomes prominent in the inner regions of the accretion flow, where the magnetic field is strong and the electrons are relativistic. The energy radiated via synchrotron depends on the strength of the magnetic field, the electron energy distribution, and the plasma temperature.

The synchrotron emissivity in an optically thin, relativistic, magnetized plasma is given by:

$$\Lambda_{syn} = \frac{16}{3}\frac{e^2}{c}\left(\frac{eB}{m_e c}\right)^2 \left(\frac{k_b T_e}{m_e c^2}\right)^2 n_e, \tag{40}$$

where B is the magnetic field strength, and $n_e = \rho/m_p$ is the electron number density.

In accretion disks dominated by toroidal magnetic fields, we express B in terms of the magnetic pressure using the plasma β parameter. Using this and the temperature relation

[5] The relation used to connect the electron temperature T_e to the proton temperature is adopted from the prescription discussed in Chattopadhyay and Chakrabarti [2002]. A detailed derivation of the bremsstrahlung and synchrotron cooling rate, including the adopted emissivity from Shapiro and Teukolsky [2008], the assumptions used for dimensional scaling, and the step-by-step transformation to the dimensionless form, is presented in Appendix D.

between electrons and protons, we derive a normalized, vertically integrated synchrotron cooling rate [Sarkar et al., 2018]:

The synchrotron cooling rate per unit volume is given by:

$$Q^-_{syn} = 1.048 \times 10^{18} \frac{e^4}{m_e^3} \frac{\beta^2}{(\beta+1)^3} \frac{\mu^2 \dot{m} a^5 \rho h}{v I_n \gamma^{5/2} x^{3/2}(x-1)} \frac{1}{2GM_\odot c} \tag{41}$$

This expression can be simplified by isolating the physical variables and grouping the constants into a coefficient, yielding:

$$\boxed{Q^-_{syn} = \frac{S \rho h a^5}{x^{3/2}(x-1)v} \frac{\beta^2}{(\beta+1)^3}} \tag{42}$$

where the S is defined as:

$$S = 1.048 \times 10^{18} \frac{e^4 \mu^2 \dot{m}}{m_e^3 I_n \gamma^{5/2}} \frac{1}{2GM_\odot c} \tag{43}$$

Here, ρ is the mass density, h is the half-thickness of the disc, a is the adiabatic sound speed, β is the plasma beta parameter, v is the radial velocity, $\dot{m}$ is the accretion rate in Eddington units, γ is the adiabatic index, I_n is a numerical factor from vertical integration.

Inverse Compton Scattering In regions where bremsstrahlung and synchrotron radiation are significant and the electron temperature is relativistically high, such as in the post-shock corona, low-energy photons can interact with energetic electrons and gain energy through inverse Compton scattering. This process enhances the energy of the outgoing radiation, leading to a harder spectrum and an increase in high-energy luminosity.

Although we do not derive a complete expression for the inverse Compton cooling rate in this work, its effects are taken into account indirectly. This is done by modifying the effective optical depth and examining how radiative efficiency varies across different regions of the accretion flow. These effects are especially important in the post-shock region, where energy transfer through Compton scattering can become a major cooling mechanism.

Since both bremsstrahlung and synchrotron cooling processes are included in the model, the total cooling rate Q^- can be expressed as the sum of the two components. Using equations 38 and 42, we obtain:

$$\boxed{Q^- = Q^-_{Br} + Q^-_{syn} = \xi \times \frac{C\rho h}{x^{3/2}(x-1)v}\left(\frac{\beta}{\beta+1}\right)^{1/2} + \frac{S\rho h a^5}{x^{3/2}(x-1)v}\frac{\beta^2}{(\beta+1)^3}} \tag{44}$$

3.3.2 Radial Advection of Toroidal Magnetic Flux

The radial advection of toroidal magnetic flux refers to the transport of the azimuthal (toroidal) component of the magnetic field, B_ϕ, through the accretion flow.

From Maxwell's equations, $\mathbf{J} = \frac{\nabla\times\mathbf{B}}{4\pi}$ and $\frac{\partial\mathbf{B}}{\partial t} = -\nabla\times\mathbf{E}$. The current density for a fluid moving with velocity $\mathbf{v}$ is given by $\mathbf{J} = \sigma(\mathbf{E}+\mathbf{v}\times\mathbf{B})$. Using these relations, the evolution equation for the magnetic field becomes:

$$\frac{\partial\mathbf{B}}{\partial t} = \nabla\times(\mathbf{v}\times\mathbf{B}) - \frac{\nabla\eta\times(\nabla\times\mathbf{B})}{4\pi}, \tag{45}$$

where $\eta = 1/\mu_0\sigma$ is the magnetic diffusivity.

The magnetic field is decomposed into mean and fluctuating components: $\mathbf{B} = (0, \langle B_\phi\rangle, 0)$ for the mean field and $\delta\mathbf{B} = (\delta B_r, \delta B_\phi, \delta B_z)$ for fluctuations, such that $B_\phi = \langle B_\phi\rangle + \delta B_\phi$. Following the study of Oda et al. [2007], the azimuthally averaged induction equation is:

$$\frac{\partial\langle B_\phi\rangle}{\partial t} = [\nabla\times(\mathbf{v}\times\langle\mathbf{B}_\phi\rangle)]_\phi + [\nabla\times\langle\delta\mathbf{v}\times\delta\mathbf{B}\rangle]_\phi + [\nabla\times(\eta\nabla\times\langle\mathbf{B}_\phi\rangle)]_\phi\,. \tag{46}$$

In the steady-state limit ($\partial\langle B_\phi\rangle/\partial t = 0$), the dynamo term (the second term on the RHS) represents toroidal field regeneration due to turbulence, and is neglected under the assumption that large-scale dynamo activity is weak. Similarly, the resistive diffusion term is ignored, assuming advection dominates over diffusion. These approximations simplify the equation to focus solely on flux transport:

$$-\frac{\partial}{\partial z}(v_z\langle B_\phi\rangle) - \frac{\partial}{\partial x}(v_x\langle B_\phi\rangle) = 0. \tag{47}$$

Assuming that the azimuthally averaged toroidal flux vanishes at the disk surface, vertical integration yields the flux advection rate $\dot{\Phi}$ as:

$$\dot{\Phi} = \int_{-\infty}^{\infty} v_x\langle B_\phi\rangle\,dz = \text{const.} \tag{48}$$

$$= -\sqrt{4\pi}\,vhB_0(x), \tag{49}$$

where

$$B_0(x) = 2^{5/4}\pi^{1/4}\left(\frac{RT}{\mu}\right)^{1/2}\Sigma^{1/2}h^{1/2}\beta^{-1/2} \tag{50}$$

is the azimuthally averaged toroidal magnetic field at the equatorial plane with R as the universal gas constant, T is the gas temperature, μ is the mean molecular weight, Σ is the vertically integrated density of the disc, h is the half-thickness of the disc, and β is the plasma beta parameter representing the ratio of gas pressure to magnetic pressure.

Based on the results of Machida et al. [2006] and Oda et al. [2007], the radial variation of the flux advection rate with distance x from the black hole is modeled as:

$$\dot{\Phi}(x;\zeta,\dot{M}) = \Phi_{\rm edge}(\dot{M})\left(\frac{x}{x_{\rm edge}}\right)^{-\zeta}, \quad \text{where} \quad \zeta = 1. \tag{51}$$

3.4 Effective Optical Depth

The transparency of the disk to radiation is characterized by the effective optical depth $\tau_{\rm eff}$, which determines whether radiation escapes freely (optically thin) or is trapped (optically thick). This parameter influences both the cooling efficiency and the emergent spectrum.

The effective optical depth combines contributions from:

- **Electron scattering opacity** τ_{es}: due to Thomson scattering of photons off free electrons.
- **Absorption opacity** τ_{abs}: from true absorption mechanisms like bremsstrahlung or synchrotron.

We use the relation for effective optical depth from [Rajesh and Mukhopadhyay, 2010]:

$$\boxed{\tau_{\rm eff} = \sqrt{\tau_{es}\tau_{abs}}} \tag{52}$$

with:

$$\tau_{es} = k_e\rho h, \quad \text{and} \quad \tau_{abs} = \frac{q_{cool}h}{4\sigma T_e^4}, \tag{53}$$

where q_{cool} is the local cooling emissivity, σ is the Stefan-Boltzmann constant, and T_e is the electron temperature.

The nature of $\tau_{\rm eff}$ determines the radiative regime:

- $\tau_{\rm eff} \ll 1$: The flow is optically thin; radiation escapes efficiently, and local cooling rates apply.

- $\tau_{\text{eff}} \gg 1$: The flow is optically thick; photons are trapped and radiative transport becomes diffusive.

In astrophysics, the concept of optical depth is crucial. It helps determine whether a disk is optically thick or thin based on the type of radiation observed. For example, optically thick disks emit radiation resembling blackbody emission, while optically thin disks radiate more energetic emissions.

4 Shocks in accretion flow

Shocks are regions where the flow properties change very rapidly and discontinuously. In the context of accretion flows, a shock wave forms when the supersonic inflow of matter is suddenly decelerated to subsonic speeds. This results in a sharp increase in density, pressure, and temperature. As matter spirals inwards, its speed increases and it experiences a stronger centrifugal force. At a certain radius, this centrifugal force can balance the gravitational pull, creating a virtual barrier. If additional matter continues to pile up against this barrier, the flow can become unstable and form shock waves. The dense region just behind the shockfront is the (PSC) Post Shock Corona region. As matter falls inwards at supersonic speeds and enters the PSC region it becomes subsonic i.e. its kinetic energy gets converted to thermal energy which in turn heats up the PSC region. Due to this shocked solutions are favored from a thermodynamic standpoint because the matter in the post-shock region attains a higher entropy state [Becker and Kazanas, 2001]. These are the regions where the flow experiences a sudden jump in properties like density magnetic pressure, etc. due to the abrupt deceleration of the inflow from supersonic to subsonic speeds.

4.1 Shock Conditions in Magnetized Accretion Flow

In steady-state, axisymmetric, vertically averaged accretion flows threaded by a toroidal magnetic field, the occurrence of a shock transition requires the satisfaction of certain conservation conditions [Landau and Lifshitz, 1987]. These conditions stem from the fundamental conservation laws of mass, momentum, magnetic flux, and energy across the discontinuity. In the presence of a toroidal magnetic field, the total pressure includes both gas and magnetic contributions.

Conservation of mass flux across the shock is given by the continuity of the vertically integrated mass accretion rate,

$$\dot{M} = 2\pi x \Sigma v,$$

where x is the cylindrical radial coordinate (in units of $r_g = 2GM/c^2$), Σ is the vertically integrated surface density, and v is the radial velocity. Continuity of this flux implies:

$$\dot{M}_- = \dot{M}_+ \tag{54}$$

$$\boxed{\Sigma_- v_- = \Sigma_+ v_+ = J} \tag{55}$$

where subscripts '–' and '+' denote quantities immediately before and after the shock, respectively, and J denotes the conserved surface mass flux.

Conservation of radial momentum ensures that the total radial flux of momentum remains continuous across the shock. The total (vertically averaged) radial momentum flux is given by:

$$\Pi = \Sigma v^2 + W, \quad \text{where} \quad W = \frac{ga^2\Sigma}{\gamma}.$$

Here, Π is the total radial momentum flux, a is the adiabatic sound speed, γ is the adiabatic index of the gas. The term W corresponds to the vertically integrated pressure. Upon dividing by the conserved mass flux J, we obtain the following compact relation:

$$\boxed{\frac{\Pi}{J} = \frac{1}{v}\left(\frac{ga^2}{\gamma} + v^2\right)} \tag{56}$$

which expresses the conserved radial momentum per unit mass flux, and is valid on both sides of the shock transition.

Conservation of magnetic flux, under the assumption of ideal MHD and a purely toroidal magnetic field, implies:

$$\dot{\Phi} = -\sqrt{4\pi} B_\phi x h = \text{constant},$$

where B_ϕ is the toroidal component of the magnetic field and h is the local half-thickness of the disk. This relation reflects the frozen-in condition of ideal MHD, whereby the magnetic flux is advected with the flow.

Using vertically averaged quantities and expressing the flux per unit mass flow, we define a conserved magnetic parameter, which remains invariant across the shock. Through algebraic manipulation[6], and using the definition of plasma beta $\beta = p_\pi/p_{\rm m}$, the conservation

[6](see Appendix G)

of toroidal magnetic flux can be written as:

$$\boxed{\frac{K}{J} = \frac{1}{\beta+1}\left(\frac{\Pi v}{J} - v^2\right)^{3/2}} \tag{57}$$

Here, Π is the vertically integrated radial momentum flux, v is the radial velocity, $J = \Sigma v$ is the surface mass flux, and β is the plasma beta parameter describing the ratio of gas pressure to magnetic pressure.

Conservation of total energy, incorporating kinetic, thermal, rotational, gravitational, and magnetic contributions, leads to:

$$\mathcal{E} = \frac{v^2}{2} + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} + \frac{a^2}{\gamma-1} + \frac{B_\phi^2}{4\pi\rho},$$

where λ is the specific angular momentum, and ρ is the density. Using the relations from above to eliminate a, B_ϕ, and ρ in terms of surface quantities and fluxes, the energy conservation condition becomes:

$$\boxed{\mathcal{E} = \frac{v^2}{2} + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} + \frac{1}{\gamma-1}\frac{\gamma}{g}\left(\frac{\Pi}{J}v - v^2\right) + \frac{K}{J}\frac{2}{v}\frac{1}{\sqrt{\frac{1}{g}\left(\frac{\Pi}{J}v - v^2\right)}}} \tag{58}$$

The four boxed conditions, equations (55)-(58), represent the necessary relations that must be simultaneously satisfied across a standing shock in a magnetized accretion flow. These equations form a closed system that can be numerically solved to determine the post-shock variables, given pre-shock conditions or vice-versa.[7]

4.2 Gradient of Flow Variables

Since we are considering steady-state hydrodynamic equations, our primary objective is to investigate the spatial variation of key flow variables within the accretion flow. In a steady-state system, time derivatives vanish, and thus, understanding the radial dependence of physical quantities becomes essential for describing the dynamics and evolution of the flow. Specifically, analyzing how variables such as velocity, density, pressure, and angular momentum vary with radial distance allows us to characterize the behavior of matter as it accretes onto the black hole.

To achieve this, we derive and examine the gradients of all relevant flow variables, which

[7] For a complete derivation of these expressions, including all algebraic steps and intermediate results, the reader is referred to Appendix G.

represent their rate of change along the radial direction. These gradients provide insight into the balance of forces and energy transport mechanisms operating in the disk. They are obtained directly from the governing equations formulated in the preceding sections and are presented as follows.

The radial gradient of the radial velocity is given by:

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{N}{D}, \tag{59}$$

where the numerator N and the denominator D are given by:

$$\begin{aligned}
N = &\frac{C(x)}{vx^{3/2}(x-1)}\frac{\beta^{1/2}}{(1+\beta)^{1/2}} + \frac{S(x)a^5}{vx^{3/2}(x-1)}\frac{\beta^2}{(1+\beta)^3} + \frac{2\alpha^2 I(p+1)\left((a^2g)+(\gamma v^2)\right)^2}{\gamma^2 xv} \\
&- \left(\frac{\lambda^2}{x^3} - \frac{1}{2(x-1)^2}\right)\left[\frac{(3+\beta(\gamma+1))v}{(\gamma-1)(1+\beta)} - \frac{4\alpha^2 gI\left((a^2g)+(\gamma v^2)\right)}{\gamma v}\right] \\
&- \frac{4\lambda\alpha I\left((a^2g)+(\gamma v^2)\right)}{\gamma x^2} - \frac{2\alpha^2 I\,(2p-3)\,(a^2g)\left((a^2g)+(\gamma v^2)\right)}{\gamma^2 xv} \\
&- \frac{2}{x-1}\left[\frac{a^2v(2+\gamma\beta)}{\gamma(\beta+1)(\gamma-1)} - \frac{2\alpha^2 I\left((a^2g)+(\gamma v^2)\right)ga^2}{\gamma^2 v}\right] \\
&+ \frac{(2p-3)}{2x}\frac{va^2}{\gamma(\gamma-1)(\beta+1)}(2\gamma\beta+3) + \frac{2(3+\beta(\gamma+1))a^2v}{\gamma(\gamma-1)(1+\beta)^2 x} \\
&- \frac{a^2v\,(4\zeta+2p-1)}{2\gamma x(\gamma-1)(1+\beta)},
\end{aligned}$$

$$\begin{aligned}
D = &\left(\frac{2a^2}{\gamma-1}\right)\left(\frac{2}{\gamma(1+\beta)} + \frac{\beta}{1+\beta}\right) - \left(\frac{(3+(\gamma+1)\beta)v^2}{(\gamma-1)(1+\beta)}\right) \\
&+ \left(\frac{2\alpha_B^2 I(a^2g+\gamma v^2)}{\gamma}\right)\left((2g-1) - \left(\frac{a^2g}{\gamma v^2}\right)\right).
\end{aligned}$$

The gradients of other key flow quantities are:

$$\frac{\mathrm{d}a}{\mathrm{d}x} = \left(\frac{a}{v} - \frac{\gamma v}{a}\right)\frac{\mathrm{d}v}{\mathrm{d}x} + \frac{\gamma}{a}\left(\frac{\lambda^2}{x^3} - \frac{1}{2(x-1)^2}\right) + \frac{a}{x-1} - \frac{(2p-3)a}{2x} - \frac{2a}{(1+\beta)x}, \tag{60}$$

$$\frac{\mathrm{d}\lambda}{\mathrm{d}x} = \frac{\alpha x\left(a^2 g - \gamma v^2\right)}{\gamma v^2}\frac{\mathrm{d}v}{\mathrm{d}x} - \frac{2\alpha a x g}{\gamma v}\frac{\mathrm{d}a}{\mathrm{d}x} - \frac{\alpha(p+1)\left(a^2 g + \gamma v^2\right)}{\gamma v}, \tag{61}$$

$$\begin{aligned}\frac{\mathrm{d}\beta}{\mathrm{d}x} = &\left(\frac{4(\beta+1)}{v} - \frac{3(\beta+1)\gamma v}{a^2}\right)\frac{\mathrm{d}v}{\mathrm{d}x} - \frac{3(\beta+1)(2p-3)}{2x} + \frac{3\gamma(\beta+1)}{a^2}\left(\frac{\lambda^2}{x^3} - \frac{1}{2(x-1)^2}\right)\\ &- \frac{6}{x} + \frac{4(1+\beta)}{x-1} + \frac{(1+\beta)\left(4\zeta + 2p - 1\right)}{2x},\end{aligned} \tag{62}$$

$$\frac{\mathrm{d}\dot{M}}{\mathrm{d}x} = p\frac{\dot{M}}{x}. \tag{63}$$

These gradient expressions form the basis for understanding how each physical variable evolves throughout the accretion disk and are crucial for identifying sonic transitions and shock formation.

4.3 Sonic Point Analysis

As matter falls toward the event horizon of a black hole, it must transition from subsonic to supersonic speeds to maintain continuity and causality. The radial location where the flow speed equals the local sound speed is termed the *sonic point* or *critical point*.

At this point, the radial velocity gradient becomes singular due to the denominator D in the expression $\frac{\mathrm{d}v}{\mathrm{d}x} = N/D$ vanishing. For the gradient to remain real and finite, the numerator N must also vanish. These two conditions, $D = 0$ and $N = 0$, are used to determine the sonic point's location and the physical values of flow variables there.

The velocity gradient $\frac{\mathrm{d}v}{\mathrm{d}x}$ at the sonic point generally yields two possible solutions, representing accretion and wind flows. When both roots are real and have opposite signs, the sonic point is of particular interest since only global transonic accretion solutions can pass through it. Such a point is classified as a saddle-type sonic point [Chakrabarti and Das, 2004]. In this work, we primarily focus on the accretion branch to study the dynamical behavior of the inflow and thus disregard wind-type solutions.

The Mach number at the sonic point, $M_c = v_c/a_c$, is determined by solving the quadratic constraint from the $D = 0$ condition:

$$M_c^2 = \frac{-m_2 - \sqrt{m_2^2 - 4m_1 m_3}}{2m_1}, \tag{64}$$

where the coefficients are given by:

$$m_1 = 2\alpha^2 I_n \gamma^2 (1+\beta_c)(\gamma-1)(2g-1) - \gamma^2(3+(\gamma+1)\beta_c), \tag{65}$$

$$m_2 = 2\gamma(2+\gamma\beta_c) + 4\alpha^2 I_n \gamma g(1+\beta_c)(g-1)(\gamma-1), \tag{66}$$

$$m_3 = -2\alpha^2 I_n g^2 (1+\beta_c)(\gamma-1). \tag{67}$$

The condition $N = 0$ leads to a fifth-order polynomial in sound speed a_c at the sonic point:

$$\mathcal{A}_c a_c^5 + \mathcal{B}_c a_c^4 + \mathcal{C}_c a_c^3 + \mathcal{D}_c a_c^2 + \mathcal{E}_c = 0, \tag{68}$$

with the coefficients defined as:

$$\mathcal{A}_c = \frac{S(x_c)}{x_c^{3/2}(x_c-1)} \cdot \frac{\beta_c^2}{(1+\beta_c)^3}, \tag{69}$$

$$\begin{aligned}\mathcal{B}_c = {} & \frac{2\alpha^2(p+1)I(g+\gamma M_c^2)^2}{\gamma^2 x_c} - \frac{2\alpha^2(2p-3)Ig(g+\gamma M_c^2)}{\gamma^2 x_c} \\ & - \frac{2}{x_c-1}\left[\frac{M_c^2(2+\gamma\beta_c)}{\gamma(\gamma-1)(1+\beta_c)} - \frac{2\alpha^2 Ig(g+\gamma M_c^2)}{\gamma^2}\right] \\ & + \frac{M_c^2(2p-3)(2\gamma\beta_c+3)}{2x_c\gamma(\gamma-1)(1+\beta_c)} + \frac{2M_c^2(3+\beta_c(\gamma+1))}{x_c\gamma(\gamma-1)(1+\beta_c)^2} \\ & - \frac{8\alpha^2 Ig(g+\gamma M_c^2)}{\gamma^2(1+\beta_c)x_c} - \frac{(4\zeta+2p-1)M_c^2}{2\gamma(\gamma-1)(1+\beta_c)x_c},\end{aligned} \tag{70}$$

$$\mathcal{C}_c = -\frac{4\lambda_c \alpha I M_c(g+\gamma M_c^2)}{\gamma x_c^2}, \tag{71}$$

$$\mathcal{D}_c = -\left(\frac{\lambda_c^2}{x_c^3} - \frac{1}{2(x_c-1)^2}\right)\left[\frac{(3+\beta_c(\gamma+1))M_c^2}{(1+\beta_c)(\gamma-1)} - \frac{4\alpha^2 gI(g+\gamma M_c^2)}{\gamma}\right], \tag{72}$$

$$\mathcal{E}_c = \frac{C(x_c)}{x_c^{3/2}(x_c-1)}\left(\frac{\beta_c}{1+\beta_c}\right)^{1/2}. \tag{73}$$

This formulation includes contributions from both bremsstrahlung and synchrotron cooling. The critical Mach number (M_c) and sound speed (a_c) determined from these conditions are used to calculate the velocity gradient$(dv/dx)_c$ at the sonic point and to integrate the flow solutions both inward and outward from this location. Now that the formulation is complete, we present results obtained by solving these equations numerically.

5 Results

5.1 Shock-free global accretion solution

With the gradient equations of all flow variables in hand, we proceed to compute the global accretion solution. The calculation begins at the inner sonic point $x_{\rm in}$, where the sound speed, velocity, and their gradients are obtained through sonic point analysis. This step requires careful treatment, as the velocity gradient exhibits *critical point behavior*, with both the numerator and denominator vanishing simultaneously. Once the solution at the sonic point is accurately determined, it is extended outward by numerically integrating the four gradient equations derived earlier. This ensures a smooth and consistent accretion profile across the entire domain.

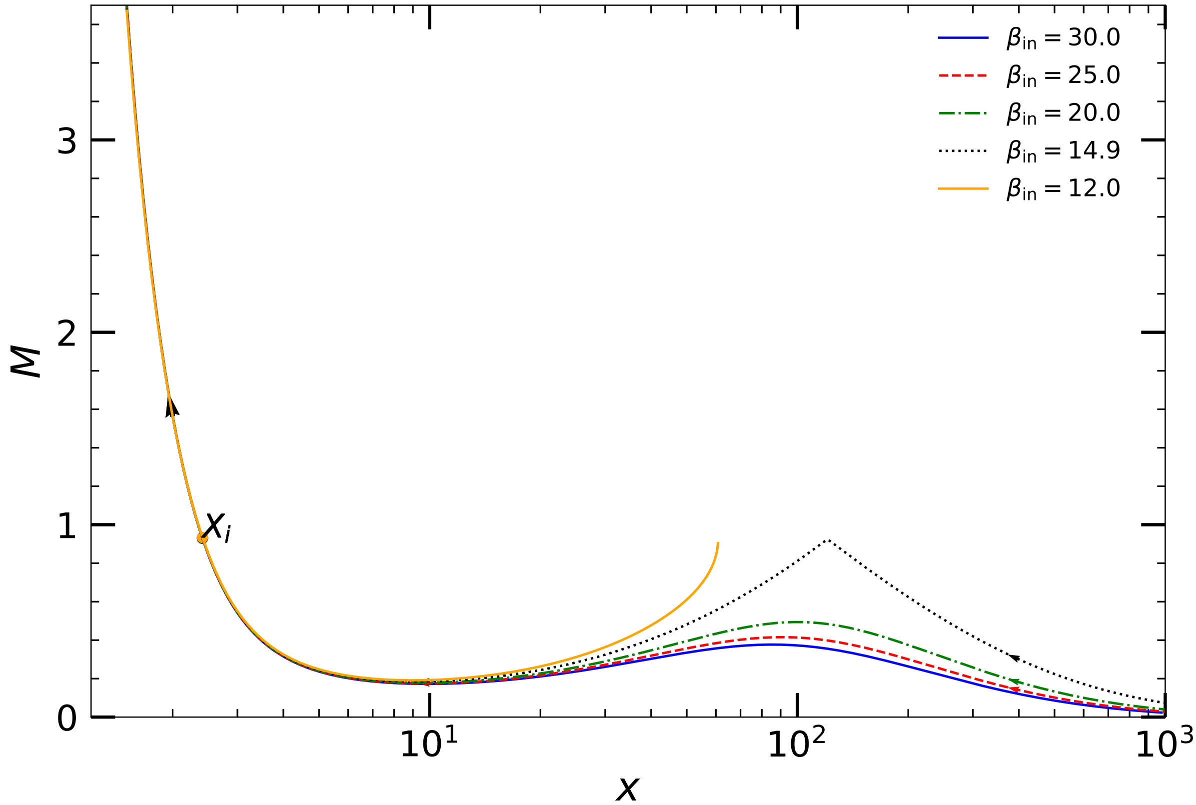


Figure 1: Illustration of radial variation of the Mach number ($M = v/a$) for accreting matter is shown for different values of the plasma parameter $\beta_{\rm in}$ at the inner sonic point $x_{\rm in} = 2.41$. Here, $\lambda_{\rm in} = 1.752$, mass loss parameter $p = 0.05$, viscosity $\alpha_B = 0.02$ bremsstrahlung cooling efficiency $\zeta = 10$, and $\dot{m}_{in} = 0.002$. Coloured curves correspond to $\beta_{\rm in} = 30$, $\beta_{\rm in} = 25$, $\beta_{\rm in} = 20$, $\beta_{\rm in} = 14.8988$ and $\beta_{\rm in} = 12$, respectively. Arrows represents the flow direction. Refer to the main text for further details.

Figure 1 displays a set of global accretion solutions corresponding to various values of the plasma parameter β_{in}, evaluated at the inner sonic point $x_{\text{in}} = 2.41$. The other flow parameters are held constant: specific angular momentum $\lambda_{\text{in}} = 1.752$, mass-loss parameter $p = 0.05$, viscosity parameter $\alpha_B = 0.02$, bremsstrahlung cooling efficiency $\xi = 10$, and accretion rate $\dot{m}_{in} = 0.002$. The colored curves represent solutions for $\beta_{\text{in}} = 30$, 25, 20, 14.8988, and 12, respectively, with arrows indicating the direction of flow.

For this configuration, the minimum value of β_{in} that allows the accretion flow to smoothly connect from the inner sonic point to the outer edge of the disk is found to be $\beta_{\text{in}}^{\text{cri}} \sim 14.8988$, shown by the black dotted curve. When $\beta_{\text{in}} < \beta_{\text{in}}^{\text{cri}}$, the accretion solution becomes incomplete and fails to extend to the outer disk. This behavior is exemplified by the solution corresponding to $\beta_{\text{in}} = 12$. All other solutions presented there except for $\beta_{\text{in}} = 12$ are known as ADAF or (Advection Dominated Accretion flow Solution) around BH.

Although the solution corresponding to $\beta_{\text{in}} = 12$ is incomplete, such configurations can still be realized by connecting them to another transonic branch that passes through the outer critical point x_o using a shock transition.

To understand the origin of this flow behavior, we examine the underlying physics in more detail. As rotating subsonic matter is injected from the outer edge of the disc, it accelerates inward due to gravitational attraction. Upon reaching the outer sonic point x_o, the flow becomes supersonic. As the matter continues its inward motion toward the black hole, it encounters a virtual barrier created by centrifugal repulsion, which causes the flow to decelerate and accumulate. This accumulation can eventually lead to a sudden transition in the flow variables, known as a shock, when the conditions for shock formation discussed in the previous section are satisfied. These shocks are thermodynamically preferred because the post-shock flow possesses higher entropy content [Becker and Kazanas, 2001].

Across the shock, flow variables such as density, plasma β, and velocity undergo abrupt changes. As the inflowing matter passes through the narrow shock region, it decelerates and converts kinetic energy into thermal energy. This leads to the formation of a dense and hot post-shock region, referred to as the post-shock corona (PSC). The subsonic matter in the PSC continues to spiral inward under the influence of strong gravity, eventually becoming supersonic again at the inner sonic point x_{in} and continuing its journey toward the event horizon.

In figure 2 we present Mach no. variation for such closed solution corresponding to $\beta_{in} = 12$ completed by incorporating a shock transition. Using the shock jump conditions outlined in the previous section, we obtain the complementary branch of the solution that passes through the outer sonic point $x_o = 190.598$, forming a fully smooth global profile Fig. 2 (shown in red). This configuration is particularly favorable, as the post-shock corona (PSC) region demonstrates elevated entropy levels, aligning with the second law of thermodynamics, which favors such high-entropy states.

Physically this figure illustrates the trajectory of subsonic matter injected at the outer edge $x_{\rm edge} = 1000$ of the disk, characterized by specific energy $\epsilon_{\rm edge} = 0.001557$, specific angular momentum $\lambda_{\rm edge} = 4.155$, plasma beta $\beta_{\rm edge} = 7.9467 \times 10^3$, bremsstrahlung cooling efficiency $\xi = 10$ and mass accretion rate $\dot{m}_{\rm edge} = 0.0027$, with a mass-loss parameter $p = 0.05$. The flow accelerates to supersonic speeds at the outer sonic point $x_o = 190.598$, then undergoes a shock transition at the centrifugal barrier located at $x_s = 33.879$, reverting to subsonic speeds. Under the influence of gravity, the flow re-accelerates to super sonic speed, passing through the inner sonic point $x_{\rm in} = 2.41$, and continues toward the black hole horizon.

Since plasma β quantifies the relative strength of gas to magnetic pressure, studying its variation helps reveal the role of magnetic fields in shaping shock dynamics within accretion flows around black holes. The next section examines this relationship in detail.

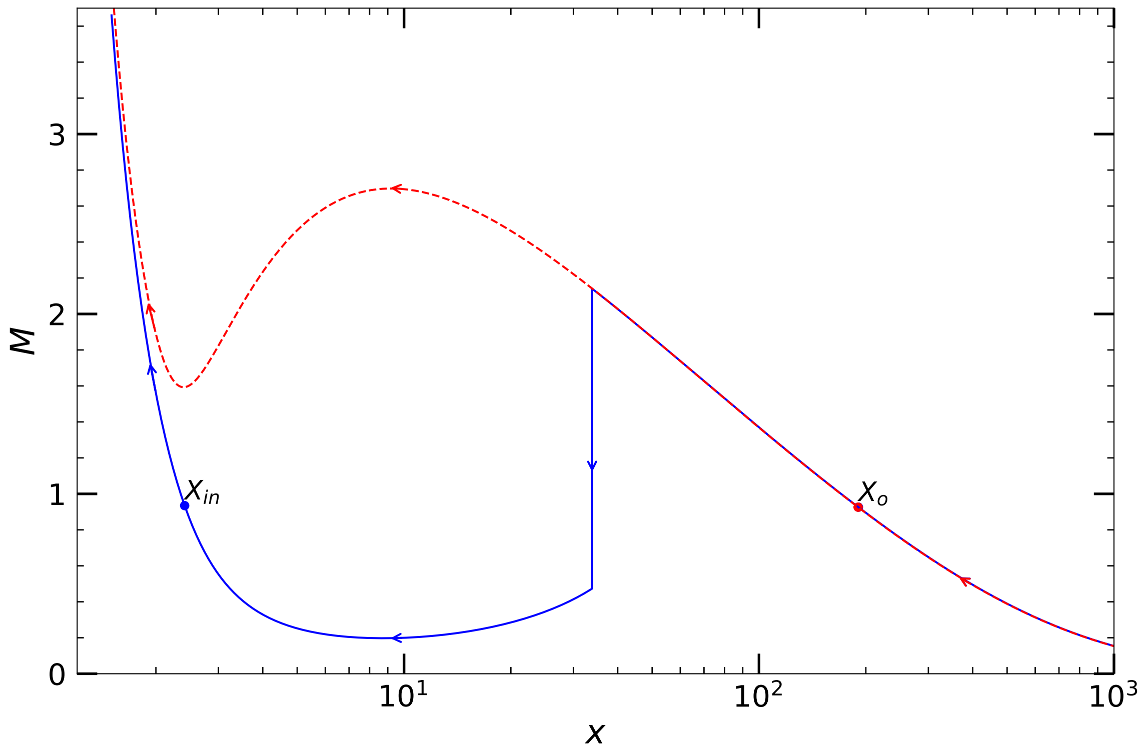
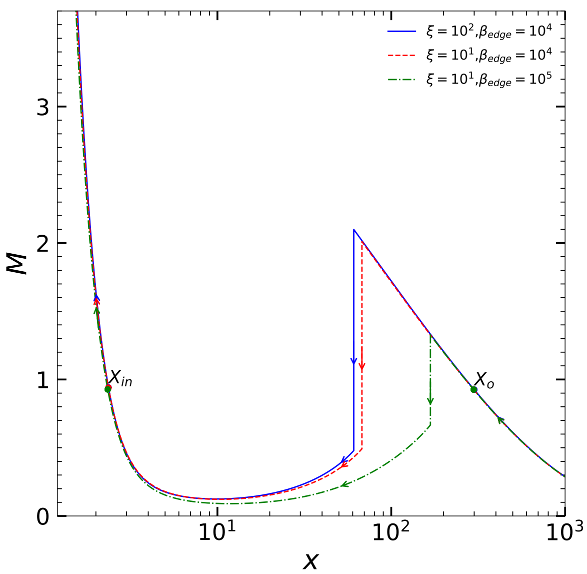
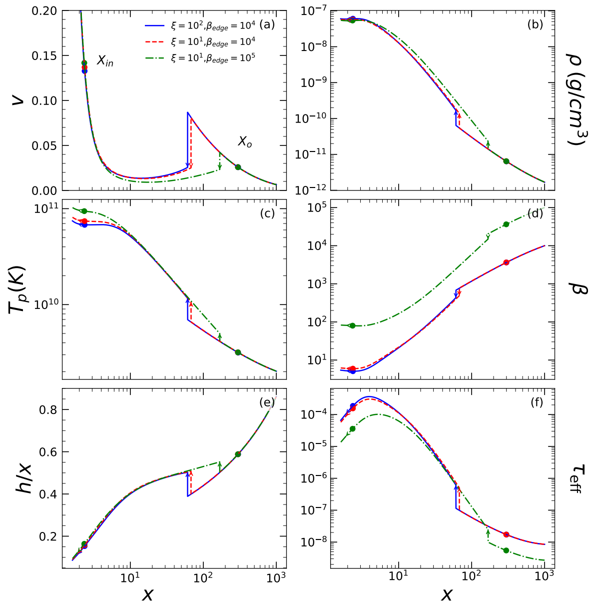


Figure 2: A comprehensive global accretion solution featuring a shock at $x_s = 33.879$ is presented, along with the locations of the outer ($x_o = 190.598$) and inner ($x_{\rm in} = 2.41$) sonic points with Mach no. ($M_o = 0.926$ and $M_i = 0.935$) respectively. The parameters used are $\beta_{\rm in} = 14.8988$, $\alpha_B = 0.02$, $\gamma = 4/3$, $\lambda_{\rm in} = 1.752$, $p = 0.05$ and $\dot{m}_{in} = 0.002$. Vertical arrow indicates the shock location and other arrows represent flow direction. Blue solid curve represents the global solution containing shock and red dashed curve is corresponding open global solution from outer edge.



(a) Variation of the Mach number as a function of the logarithmic radial distance is shown for different values of the plasma beta parameter β_{edge} and bremsstrahlung cooling efficiency ξ. The inflowing matter is injected at subsonic speed from the outer edge at $x_{\text{edge}} = 1000$ with specific energy $\epsilon_{\text{edge}} = 1.0109 \times 10^{-3}$, specific angular momentum $\lambda_{\text{edge}} = 3.018$, mass-loss parameter $p = 0.05$, accretion rate $\dot{m}_{\text{edge}} = 0.01$, and viscosity parameter $\alpha_B = 0.02$. We examine shock solutions for three cases: (i) $\beta_{\text{edge}} = 10^4$, $\xi = 100$ (blue solid line), (ii) $\beta_{\text{edge}} = 10^4$, $\xi = 10$ (red dashed line), and (iii) $\beta_{\text{edge}} = 10^5$, $\xi = 10$ (green dash-dotted line). The vertical arrows indicate the locations of the corresponding shocks, which shift inwards with decreasing β_{edge} and increasing ξ, while all other arrows denote the flow direction.



(b) The figure illustrates the variation of radial velocity v, density ρ (in g/cm^3), proton temperature T_p in Kelvin, the ratio of gas pressure to magnetic pressure (β), disk scale height (h/x), and effective optical depth (τ_{eff}) as a function of $\log(x)$ for a stellar mass BH of mass $M_{BH} = 10 M_{\odot}$. The inner and outer sonic points (x_{in}, x_{out}) are marked with circles. The vertical arrow indicates the shock location, while the other arrows represent the flow direction. We examine shock solutions for three cases: (i) $\xi = 100$, βedge $= 10^4$ (blue solid line), (ii) $\xi = 10$, βedge $= 10^4$ (red dashed line), and (iii) $\xi = 10$, βedge $= 10^5$ (green dash-dotted line). The outer edge flow parameters used in all cases are $\epsilon_{\text{edge}} = 1.0109 \times 10^{-3}$, $\lambda_{\text{edge}} = 3.018$, $\dot{m}_{\text{edge}} = 0.001$, $p = 0.05$, $\alpha_B = 0.02$, and $\gamma = 4/3$.

Figure 3: Shock structure and flow properties for varying plasma beta β_{edge} and bremsstrahlung cooling efficiency ξ.

5.2 Shocks dynamics with β

In Figure 3a, we show the shock solutions for subsonic matter injected at the outer edge with parameters: $\epsilon_{\text{edge}} = 1.0109 \times 10^{-3}$, $\lambda_{\text{edge}} = 3.018$, $\beta_{\text{edge}} = 10^5$, $\xi = 10$, $\dot{m}_{\text{edge}} = 0.01$, and $p = 0.05$. Under these conditions, the flow undergoes a shock transition at $x_s = 167.99$. Next, we reduce the plasma beta to $\beta_{\text{edge}} = 10^4$ while keeping all other parameters the same. This causes the shock front to move inward, forming at $x_s = 67.82$. A lower β_{edge} enhances synchrotron cooling and increases magnetic turbulence. This strengthens the Maxwell stress, promoting more efficient angular momentum transport from the inner to the outer disc. As a result, the centrifugal barrier in the post-shock region becomes weaker, and the shock front moves closer to the black hole.

Finally, we increase the bremsstrahlung cooling efficiency to $\xi = 100$ for $\beta_{\text{edge}} = 10^4$, keeping all other outer edge parameters unchanged. This causes the shock to move even further inward to $x_s = 60.87$. These results highlight the significant effect of thermal pressure on shock dynamics. As cooling increases, thermal pressure in the post-shock region decreases, leading to a new force balance closer to the horizon.

In Fig. 3b we study the variation of flow variables to understand the disk structure further for 3 cases from Fig. 3a for a stellar mass BH of mass $M_{BH} = 10M_\odot$. In each panel we have plotted particular flow property to get better insights of flow. Fig. 3b (a) illustrates the variation of radial velocity (v) with distance from the black hole. As the flow moves inward, v increases steadily until it encounters a shock, where it experiences a sudden drop to subsonic speeds. Beyond the shock, the velocity rises once more in the inner region. After crossing the inner critical point, the flow accelerates rapidly and enters the black hole at relativistic speeds.

In Fig. 3b (b), we show the radial variation of the density. As the flow approaches the shock, it slows down due to the centrifugal barrier, causing matter to pile up near the shock location. This leads to a sharp rise in density. Beyond the shock, the density continues to rise smoothly as the flow moves inward. Additionally, it is evident that with stronger cooling, the rise in density near the shock becomes sharper. This behaviour can be attributed to the reduction in thermal pressure, as lower pressure allows matter to compress more efficiently, leading to a sharper density enhancement.

In Fig. 3b (c), we illustrate the variation of proton temperature (T_p) throughout the accretion flow. At the shock location, the supersonic inflow undergoes a sudden transition to subsonic speeds, resulting in the conversion of kinetic energy into thermal energy. This leads to a significant rise in temperature in the post-shock region. Furthermore, we observe that lowering the value of $\beta_{\rm edge}$ yields an overall cooler temperature profile across the disc, suggesting that increased magnetic pressure suppresses thermal heating. This trend is consistent with numerical simulations [Machida et al., 2006], which show that magnetically dominated accretion flows tend to form cooler disc structures. Finally, with higher bremsstrahlung cooling efficiency ξ, the temperature further decreases, particularly near the inner region of the disc, due to more effective radiative cooling.

In Fig. 3b (d), we display the radial distribution of the plasma β parameter. As the accretion flow advances towards the black hole, β steadily decreases, indicating a growing dominance of magnetic pressure. Notably, across the shock, β undergoes a rapid drop, signifying that the post-shock region becomes magnetically dominated. In Fig. 3b (e), we show the radial profile of the half thickness to radius ratio (h/x). Throughout the flow, including near the outer edge, this ratio remains well below unity, thereby supporting the validity of our thin disc approximation. In Fig. 3b (f), we depict the radial variation of the effective optical depth, defined as $\tau_{\rm eff} = \sqrt{\tau_{\rm es}\tau_{\rm cool}}$. Here, $\tau_{\rm es}$ corresponds to the electron scattering optical depth and is calculated using $\tau_{\rm es} = \kappa_{\rm es}\rho h$, where $\kappa_{\rm es} = 0.38$ cm^2 g^{-1} represents the electron scattering opacity. The absorptive optical depth due to radiative cooling, $\tau_{\rm cool}$, is estimated from $\tau_{\rm cool} = hq_{\rm cool}/4\sigma T_e^4 \times 2GM_{\rm BH}/c^2$, where $q_{\rm cool} = q_{\rm syn} + q_{\rm brm}$ is the local cooling emissivity and σ is the Stefan-Boltzmann constant and $M_{BH} = 10M_\odot$. We observe that the effective optical depth is significantly enhanced within the post-shock region due to the density rise there (refer to Fig. 3b (b)). Additionally, for higher values of accretion rate, the optical depth tends to remain elevated throughout the disc. Despite this, the post-shock region stays optically thin ($\tau_{\rm eff} < 1$), implying that high-energy radiation from this zone can still escape efficiently.

5.2.1 Shock Characteristics

Variation of shock characteristics with respect to bremsstrahlung cooling factor ξ.

We now investigate how the shock characteristics of the accretion flow are influenced by variations in the cooling efficiency parameter ξ. In Fig. 4, we present the variation of (a) logarithmic shock location x_s, (b) compression ratio R, and (c) shock strength Θ for flows launched from $x_{\rm edge} = 1000$, with fixed values of $\beta_{\rm edge} = 5.87 \times 10^4$, $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, $\lambda_{\rm edge} = 3.01837$, and mass loss parameter $p = 0.05$. The results are plotted as a function of ξ, and the solid, dotted, and dashed curves correspond to accretion rates $\dot{m}_{\rm edge} = 0.03$, 0.02, and 0.01, respectively.

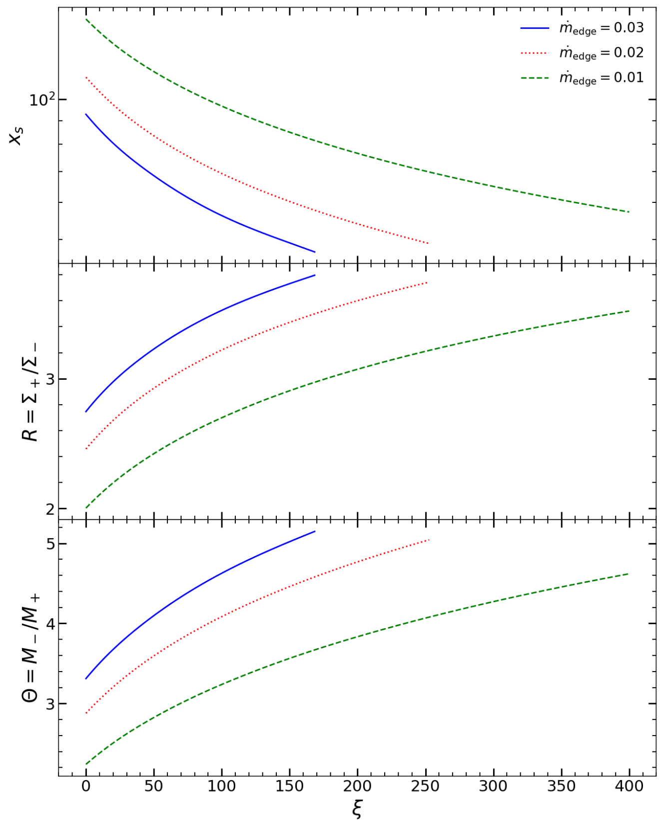


Figure 4: Variation of (a) logarithmic shock location x_s, (b) compression ratio R, and (c) shock strength Θ for flows injected from $x_{\rm edge} = 1000$ with $\beta_{\rm edge} = 5.87 \times 10^4$, $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, $\lambda_{\rm edge} = 3.01837$, and mass-loss parameter $p = 0.05$; all as a function of ξ. The solid, dotted, and dashed curves represent the results for $\dot{m}_{\rm edge} = 0.03$, 0.02, and 0.01, respectively. Further details are provided in the text.

From the figure, it is evident that stationary shocks can exist over a broad range of ξ. For a given mass accretion rate $\dot{m}_{edge}$ at the outer edge, increasing the bremsstrahlung cooling efficiency ξ causes the shock front to move inward, closer to the black hole horizon. This inward shift arises due to enhanced radiative losses, which reduce the energy content of the flow as it accretes. Additionally, we observe that increasing the mass accretion rate $\dot{m}_{\rm edge}$ also causes the shock front to shift inward. This is because a higher accretion rate amplifies both bremsstrahlung and synchrotron cooling, which lowers the thermal pressure in the post-shock region (PSC), thereby allowing the shock to settle at a smaller radius.

As ξ increases beyond a certain critical value, the conditions required for a standing shock are no longer satisfied, and the shock disappears altogether. This suggests that strong cooling acts against the formation of steady shocks, reducing their likelihood. The critical cooling parameter beyond which shocks vanish depends sensitively on the initial conditions of the flow at the outer boundary.

In addition to the shock location, we also present the variation of the shock compression ratio and shock strength for all cases. It is observed that both quantities increase as the shock front shifts closer to the black hole horizon. This behavior arises because, at smaller radii, the infalling matter possesses higher kinetic energy due to the stronger gravitational pull. Consequently, the pre-shock Mach number increases, leading to a stronger discontinuity at the shock front. As a result, the post-shock region becomes denser and hotter, which enhances both the compression ratio and the shock strength.

Variation of shock characteristics with respect to $\beta_{\rm edge}$.

Next, we explore how the shock properties of the accretion flow are influenced by variations in two key parameters: the plasma parameter β and the viscosity parameter α_B. In Fig. 5, we present a set of shock solutions for flows launched from $x_{\rm edge} = 1000$, with fixed parameters: mass-loss fraction $p = 0.05$, specific energy $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, angular momentum $\lambda_{\rm edge} = 3.0189$, and accretion rate $\dot{m}_{\rm edge} = 0.01$. The results are shown as a function of $\beta_{\rm edge}$, where the solid, dotted, and dashed curves correspond to viscosity values $\alpha_B = 0.0210$, 0.0205, and 0.0200, respectively.

By varying $\beta_{\rm edge}$, we examine its effect on the shock structure. In the first panel of Fig. 5, we illustrate the behavior of the shock location as a function of $\beta_{\rm edge}$. It is evident that lowering $\beta_{\rm edge}$ causes the shock front to shift inward, closer to the black hole horizon. This occurs because a smaller β value corresponds to a stronger magnetic field, which enhances synchrotron cooling process. The combined effect of increased cooling reduces the thermal pressure in the post-shock region (PSC), thereby weakening the pressure support needed to sustain a shock at larger radii, and consequently driving the shock closer to the black hole.

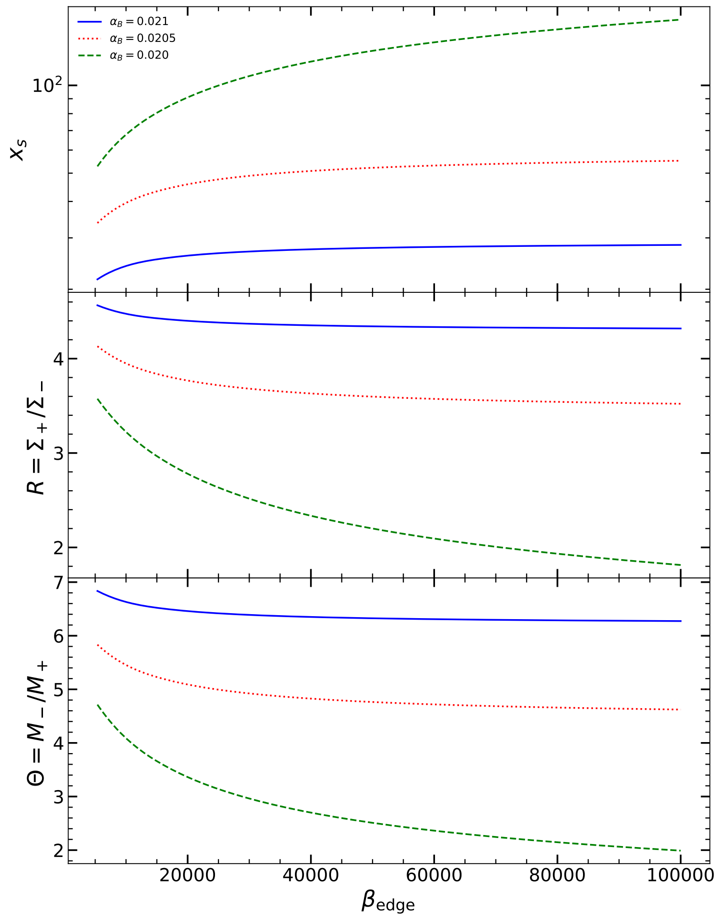


Figure 5: Variation of (a) logarithmic shock location x_s, (b) compression ratio R, and (c) shock strength Θ for flows injected from $x_{\rm edge} = 1000$ with mass-loss parameter $p = 0.05$, $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, and $\dot{m}_{\rm edge} = 0.01$; all as a function of $\beta_{\rm edge}$. The solid, dotted, and dashed curves represent the results for $\alpha_B = 0.0210$, 0.0205, and 0.0200, respectively. Further details are provided in the text.

Variation of Shock characteristics with α_B

Additionally, we observe that increasing the viscosity parameter α_B also causes the shock front to shift inward. This behavior can be understood as follows: as α_B increases, viscous forces within the flow become stronger, leading to more efficient outward transport of angular momentum. As a result, the centrifugal barrier that opposes gravity comes closer to the BH. Consequently, the shock forms at a smaller radial distance. Moreover, higher viscosity also leads to stronger shocks, as the inward-shifted shock interacts with infalling matter carrying higher kinetic energy. This results in an increased compression ratio and greater shock strength.

5.3 Mass loss parameter

In radiatively inefficient accretion flows, a large portion of the inflowing gas may be expelled through winds before reaching the black hole. This behavior is described by the mass-loss parameter p, which governs the radial variation of the accretion rate as $\dot{M}(x) \propto x^p$, with $0 \leq p < 1$. Blandford and Begelman [1999] introduced the Advection-Dominated Inflow-Outflow Solution (ADIOS) as an extension of the ADAF model, emphasizing that only a small fraction of the supplied mass is accreted, while the rest escapes as a wind, carrying away energy and angular momentum. Increasing p strengthens this outflow, thereby modifying the disc structure and reducing the accretion rate near the black hole.

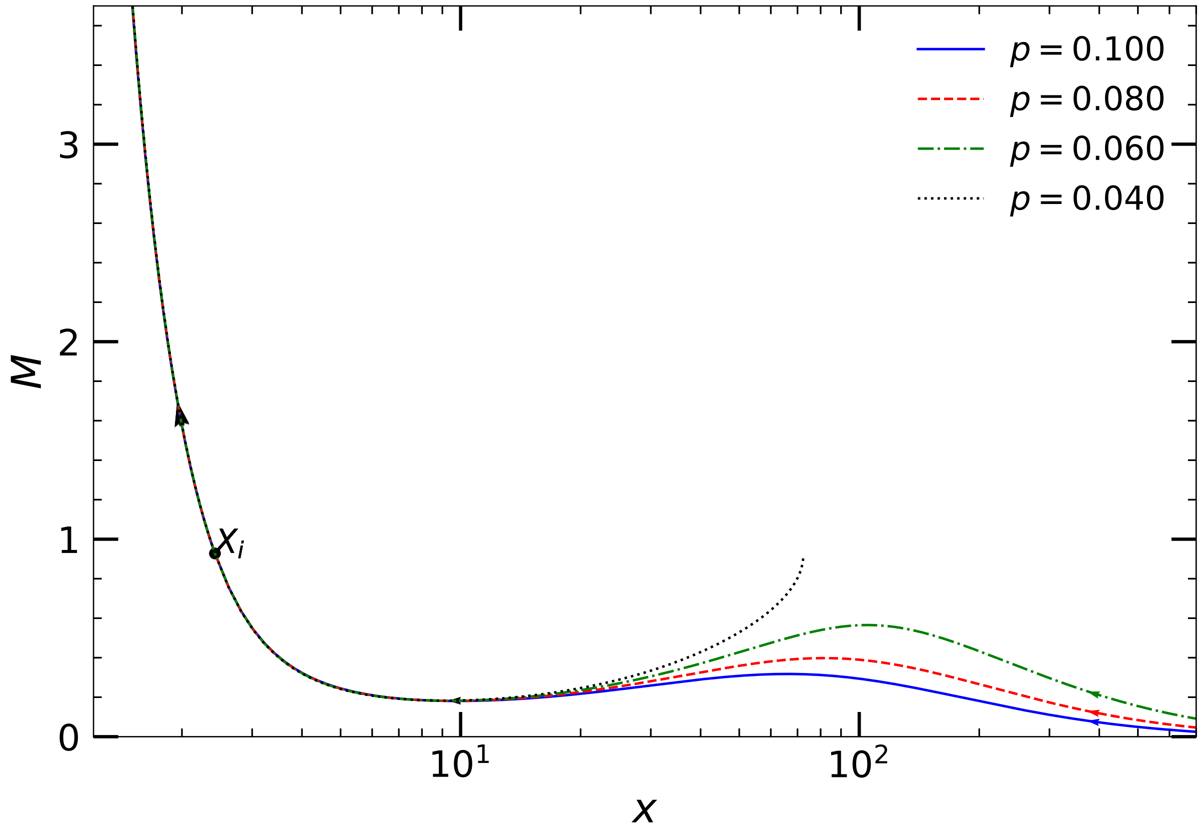


Figure 6: Illustration of radial variation of the Mach number ($M = v/a$) for accreting matter is shown for different values of the mass-loss parameter, $\dot{m}_{in} = 0.001$ at the inner sonic point $X_{\rm i} = 2.42$. Here, $\lambda_{\rm in} = 1.75$, $\alpha_B = 0.02$, and $\beta_{in} = 50$. The colored curves with different line styles correspond to different values of p, $p = 0.10$ (solid blue), $p = 0.08$ (dashed red), $p = 0.06$ (dash-dotted green) and (dotted black) $p = 0.04$. Where we took both bremsstrahlung cooling (with cooling factor $\xi = 10$) under consideration. Arrows represent the flow direction.

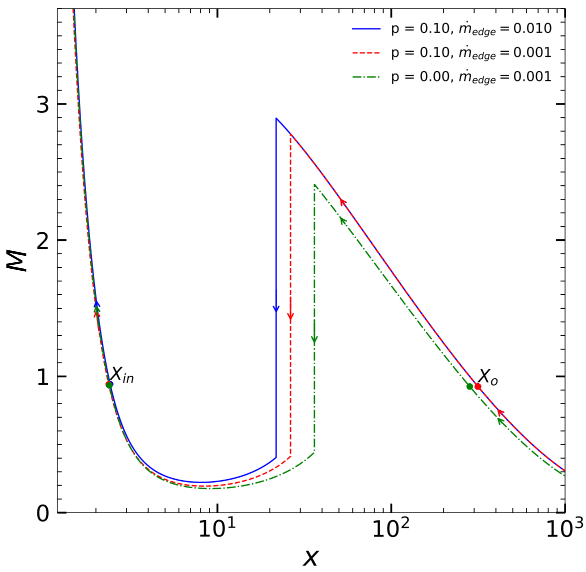
To incorporate this effect, we adopt the modified accretion rate formulation given in equation 6. We investigate its impact by computing global transonic solutions starting from the inner sonic point $x_{\rm in} = 2.42$, with $\lambda_{\rm in} = 1.75$, $\dot{m}_{in} = 0.001$, $\alpha_B = 0.02$, and $\beta_{\rm in} = 50$, including bremsstrahlung cooling with $\xi = 10$. The curves corresponding to $p = 0.10$ (solid blue), 0.08 (dashed red), 0.06 (dash-dotted green), and (dotted black) $p = 0.04$. Figure 6 illustrate how increasing mass-loss parameter alters the global accretion profile, with higher p leading to more open flow structures.

5.4 Shock dynamics with mass loss parameter

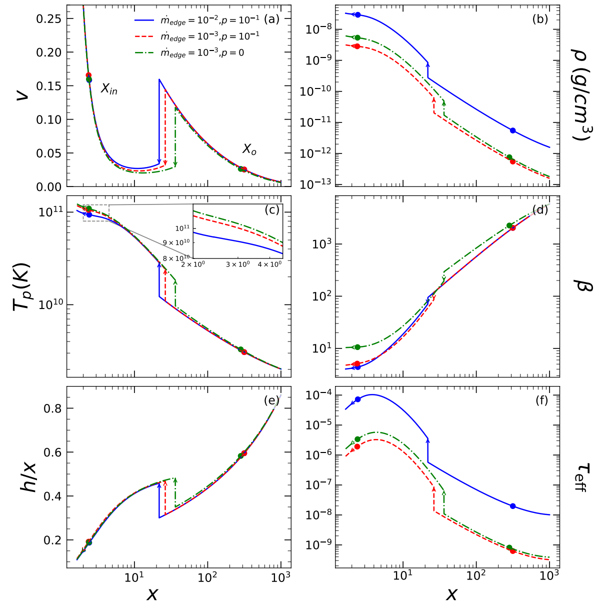
In this section, we analyze how the shock dynamics in an accretion flow are affected by variations in the mass-loss parameter p and the mass accretion rate $\dot{m}_{\rm edge}$. We consider global accretion solutions for flows launched from the outer edge at $x_{\rm edge} = 1000$, characterized by specific energy $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, plasma beta parameter $\beta_{\rm edge} = 5.87 \times 10^4$, specific angular momentum $\lambda_{\rm edge} = 2.968$, and viscosity $\alpha_B = 0.02$. First we take the case with mass accretion rate $\dot{m}_{edge} = 0.001$ and mass loss parameter $p = 0.00$ which gives a standing shock at $x_S = 36.14$, then keeping all other parameters same we increase the value of mass loss parameter to $p = 0.10$ which pushed the shock inwards to $x_s = 26.34$. The results show that as the mass-loss parameter p increases, the shock front shifts inward toward the black hole. This behavior arises due to the reduction in the effective mass inflow rate at smaller radii, caused by mass loss through winds. More value of p means more outflows and winds which leads to less mass being accumulated in post-shock region leading to decreased pressure, thereby weakening the resistance against the gravitational pull of the central compact object. As a result, the shock front moves to smaller radii with increasing p, demonstrating a direct correlation between the strength of mass outflow and the radial position of the standing shock. Furthermore, the figure also highlights the influence of accretion rate: keeping all other parameters same and increasing mass accretion rate to $\dot{m}_{\rm edge} = 0.01$ also moves the shock front inward to $x_s = 21.76$. Although this might appear contradictory at first, it is explained by the fact that a higher accretion rate enhances the overall cooling rate in the flow. Increased radiative cooling reduces thermal pressure in the PSC, which again shifts the shock location closer to the black hole. Thus, both mass-loss and increased accretion rate act to push the shock inward through different but complementary mechanisms.

To further investigate the impact of the mass-loss parameter p on shock dynamics and accretion flow, we analyze the variation of key flow variables for a stellar mass BH of mass $M_{BH} = 10 M_\odot$, as depicted in Figure 7b. Panel (a) shows the radial velocity profile $v(x)$, illustrating how subsonic material injected from the outer edge accelerates and undergoes a shock transition before accreting onto the black hole.

In panel (b), we present the density profile $\rho(x)$ corresponding to the cases shown in Figure 7a. It is evident that, for the same outer-edge accretion rate $\dot{m}_{\rm edge}$, the flow with a higher mass-loss parameter ($p = 0.1$) exhibits lower density compared to the $p = 0$ case. This reflects the reduction in mass due to winds and outflows. Additionally, increasing



(a) Variation of the Mach number as a function of the logarithmic radial distance is shown for different values of the mass-loss parameter p. The inflowing matter is injected at subsonic speed from the outer edge at $x_{\rm edge} = 1000$ with specific energy $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, plasma beta parameter $\beta_{\rm edge} = 5.87{\times}10^4$, specific angular momentum $\lambda_{\rm edge} = 2.968$, and viscosity parameter $\alpha_B = 0.02$. We examine shock solutions for three cases: (i) $p = 0.10$ with $\dot{m}_{\rm edge} = 0.01$ (blue solid line), (ii) $p = 0.10$ with $\dot{m}_{\rm edge} = 0.001$ (red dashed line), and (iii) $p = 0.00$ with $\dot{m}_{\rm edge} = 0.001$ (green dash-dotted line). The vertical arrows indicate the locations of the corresponding shocks, which shift inwards with increasing p, while all other arrows denote the flow direction.



(b) The figure illustrates the variation of radial velocity v, density ρ (in g/cm^3), proton temperature T_p in Kelvin, the ratio of gas pressure to magnetic pressure, disk scale height (h/x), and effective optical depth $(\tau_{\rm eff})$ as a function of $\log(x)$ for a stellar mass BH of mass $M_{BH} = 10 M_\odot$. The inner and outer sonic points ($x_{\rm in}$, $x_{\rm out}$) are marked with circles. The vertical arrow indicates the shock location, while the other arrows represent the flow direction. We examine shock solutions for three cases: (i) $p = 0.10$ with $\dot{m}_{\rm edge} = 0.01$ (blue solid line), (ii) $p = 0.10$ with $\dot{m}_{\rm edge} = 0.001$ (red dashed line), and (iii) $p = 0.00$ with $\dot{m}_{\rm edge} = 0.001$ (green dash-dotted line). The flow parameters used include $\beta_{\rm edge} = 5.87 \times 10^4$, $\alpha_B = 0.02$, $\gamma = 4/3$, and $\lambda_{\rm edge} = 2.968$.

Figure 7: Shock location and flow variable profiles for different mass-loss parameters p

$\dot{m}_{\rm edge}$ results in a higher density throughout the flow, as expected from the initially larger mass inflow.

Panel (c) of Figure 7b shows the variation of proton temperature T_p. All three cases exhibit a sharp increase in temperature within the post-shock region (PSC), indicating the heating caused by the shock transition. As discussed earlier, an increase in $\dot{m}_{\rm edge}$ shifts the shock closer to the black hole. This is further confirmed here, as the case with the highest accretion rate shows the lowest T_p in the PSC, highlighting the effect of enhanced cooling and reduced thermal pressure. Furthermore, when keeping $\dot{m}_{\rm edge}$ fixed, increasing p results in a noticeable decrease in T_p within the PSC. This reduction in temperature reflects a drop in thermal pressure due to mass loss, which in turn drives the shock front inward to re-establish pressure balance.

Panel (d) of Figure 7b shows the radial variation of the plasma parameter β for the solutions presented in Figure 7a. As the flow accretes, β decreases, with a sharp drop at the shock location, indicating a sudden rise in magnetic pressure in the post-shock region (PSC). For a fixed $\dot{m}_{\rm edge}$, a higher mass-loss parameter ($p = 0.1$) results in a steeper decline in β, implying enhanced magnetic turbulence and more efficient outward angular momentum transport. This indicates that the flow becomes magnetically dominated near the black hole. In contrast, the case with $p = 0$ shows a more gradual decrease in β, suggesting that thermal pressure continues to dominate in the inner region.This further supports our theoretical prediction that an increase in the mass-loss parameter leads to the inward movement of the shock front. The enhanced magnetic dominance and reduced thermal pressure in the inner region due to mass loss contribute to this inward shift, thereby validating the physical mechanism driving the shock dynamics.

In Panel (e), we present the variation of the disk half-thickness to radial distance ratio, which remains well below unity throughout the flow. This supports the validity of our thin disk approximation. Panel (f) displays the optical depth profile for the same solutions, which also stays significantly less than one. Despite the sharp increase in density near the shock, the optical depth remains low ($\tau_{\rm eff} < 1$), thereby justifying the assumption of an optically thin accretion disk in our model.

Variation of shock characteristics with respect to the mass-loss parameter p.

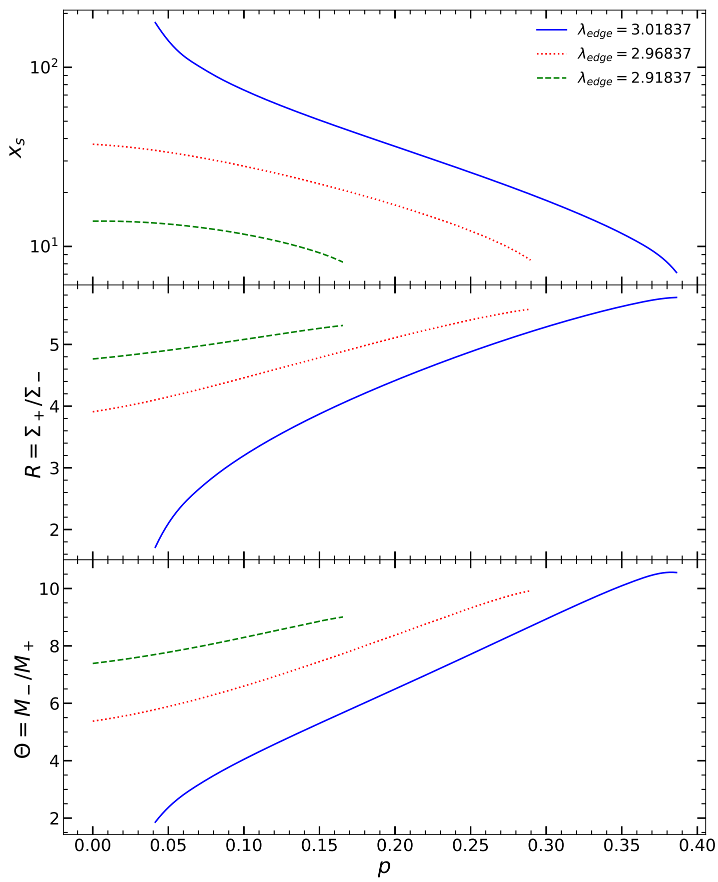


Figure 8: Variation of (a) logarithmic shock location x_s, (b) compression ratio R, and (c) shock strength Θ for flows injected from $x_{\rm edge} = 1000$ with $\beta_{\rm edge} = 5.87 \times 10^4$, $\epsilon_{\rm edge} = 1.0109 \times 10^{-3}$, and $\dot{m}_{\rm edge} = 0.01$; all as a function of the mass-loss parameter p. The solid, dotted, and dashed curves represent the results for $\lambda_{\rm edge} = 3.01837$, 2.96837, and 2.91837, respectively. Further details are provided in the text.

Now that we understand the impact of the mass-loss parameter on shock dynamics, we proceed to explore how shock characteristics vary with changes in both the mass-loss parameter and the specific angular momentum of the flow. In Fig. 8, we present a set of solutions for subsonic matter injected from the outer edge at $x_{\text{edge}} = 1000$, with $\beta_{\text{edge}} = 5.87 \times 10^4$, $\epsilon_{\text{edge}} = 1.0109 \times 10^{-3}$, and $\dot{m}_{\text{edge}} = 0.01$. We consider three different values of specific angular momentum: $\lambda_{\text{edge}} = 3.01837$, 2.96837, and 2.91837.

We then vary the mass-loss parameter p across a range and examine how the (a) logarithmic shock location x_s, (b) shock compression ratio R, and (c) shock strength Θ evolve as a function of p. From Fig. 8, it is evident that as the mass-loss parameter p increases, the shock position moves closer to the black hole horizon for all three cases of specific angular momentum. However, for a given set of flow parameters, p cannot be increased indefinitely. Beyond a certain critical value p_{crit}, the necessary conditions for shock formation are no longer satisfied, and standing shocks cease to exist. Other than that, we also present the variation of (b) compression ratio R and (c) shock strength Θ for the flow. These results consistently show that the closer the shock is to the black hole horizon, the higher the compression ratio. This is because matter encountering the shock at smaller radii possesses higher kinetic energy, leading to stronger compression and a more intensified shock. Finally, it is also evident that increasing the specific angular momentum λ_{edge} shifts the shock front outward, confirming that the shocks under consideration are centrifugally driven.

6 Conclusion

In this project, we conducted a detailed investigation of magnetized accretion flows around a Schwarzschild black hole using the magnetohydrodynamic (MHD) formalism. The focus was on understanding the role of toroidal magnetic fields, radiative cooling mechanisms, viscous dissipation, and shock transitions in shaping the physical and dynamical properties of the accretion disk. The work integrates theory, mathematical modeling, and numerical computation to explore the interplay between plasma physics and gravity in black hole environments.

We began by discussing the fundamental aspects of accretion onto compact objects, emphasizing the importance of magnetic fields in the dynamics of the accreting plasma. Adopting a toroidal magnetic field configuration, which naturally emerges due to the differential rotation of the disk, we constructed an MHD model suitable for geometrically thin, optically thin, axisymmetric accretion disks. The validity of applying the MHD framework was justified through the evaluation of the *Knudsen number* and *Larmor* *number*. The governing equations included the continuity equation, radial and azimuthal momentum equations, entropy generation equation, and the evolution of the toroidal magnetic flux. These were derived under steady-state conditions and incorporated the pseudo-Newtonian Paczyński-Wiita potential to mimic general relativistic effects near the event horizon.

To model radiative losses, we included both bremsstrahlung and synchrotron cooling mechanisms. The dimensionless forms of the cooling rates were derived and expressed in terms of physical quantities like density, pressure, and sound speed. We also computed the effective optical depth using the absorption and scattering contributions, showing that the disk remains optically thin throughout, even in the dense post-shock regions. This consistency with the optically thin assumption provides confidence in the reliability of the radiative treatment used.

We examined the possibility of standing shocks in the accretion flow by applying the Rankine-Hugoniot conditions along with magnetic flux conservation at the shock front. The presence of shocks was found to be strongly dependent on the physical parameters at the outer boundary, including plasma beta β, viscosity parameter α_B, cooling efficiency ξ, and the mass-loss parameter p. For a wide range of parameters, we obtained global transonic solutions containing standing shocks, which represent a discontinuous transition from supersonic to subsonic motion. These shocks significantly modify the flow by increasing the density, temperature, and magnetic pressure in the post-shock corona (PSC), thereby impacting the emission characteristics of the disk.

We performed a parametric study to understand the sensitivity of the shock structure to different flow parameters. It was observed that decreasing the plasma beta β, which corresponds to increasing magnetic pressure relative to gas pressure, pushes the shock front inward. This occurs because stronger magnetic fields enhance synchrotron cooling and suppress thermal pressure in the PSC, reducing its ability to withstand gravitational infall. A similar inward shift of the shock location was observed when the viscosity parameter α_B was increased. Enhanced viscosity facilitates the outward transport of angular momentum, which leads to the centrifugal barrier moving closer to the black hole, thus favoring shock formation at smaller radii.

Additionally, increasing the bremsstrahlung cooling efficiency ξ resulted in the shock moving inward as well. This is because stronger cooling reduces the thermal pressure support in the PSC, allowing gravity to dominate more effectively. When the cooling strength exceeds a critical value, the standing shock ceases to exist as the conditions for its formation are no longer satisfied. This points to a threshold in radiative dissipation beyond which shocks cannot form in steady-state solutions. Similarly, we demonstrated that a higher mass accretion rate $\dot{m}$ enhances both cooling mechanisms, contributing to an inward migration of the shock front.

We also studied the effects of mass-loss through outflows using the parameter p. Increasing p reduces the effective inflow rate into the PSC, leading to lower pressure and a consequent inward shift of the shock location. However, this trend is limited by a critical value p_{crit} beyond which standing shocks are no longer supported. This behavior illustrates the strong influence of outflows in modifying the inner disk structure and thermodynamics.

Shock characteristics such as shock strength Θ and compression ratio R were also analyzed. We found that as the shock forms closer to the black hole, both R and Θ increase. This is because the infalling matter carries higher kinetic energy near the black hole, resulting in stronger shocks and compression. Moreover, the variation of these quantities with different angular momentum values revealed that shocks move outward as the angular momentum at the outer edge increases, reaffirming their centrifugal origin.

The post-shock region was further analyzed in terms of radial velocity, density, temperature, plasma beta, disk height-to-radius ratio, and optical depth. The temperature and density exhibited strong jumps across the shock, indicating significant energy dissipation. The plasma β dropped sharply, reflecting magnetic domination in the PSC. Despite the density increase, the optical depth remained well below unity, confirming the optically thin nature of the disk. The disk thickness to radius ratio remained under unity throughout, supporting the geometrically thin approximation used in our model.

In summary, this project has provided an in-depth exploration of magnetized, radiatively cooled accretion flows around black holes, emphasizing the significance of toroidal magnetic fields and shock formation. By numerically solving the governing equations and investigating the effects of key physical parameters, we have demonstrated the conditions under which standing shocks occur and how they modify the accretion dynamics. The outcomes have significant implications for understanding high-energy emissions from black hole systems, as the PSC is expected to be a major source of X-rays. The framework and findings of this study can be extended further to incorporate more complex physics, including two-temperature flows, general relativistic effects, and self-consistent treatment of jets and winds.

Appendices

A Lorentz Transformation of Electromagnetic Fields

To describe fields in different frames, consider a fluid moving with velocity $\mathbf{u}$. The electromagnetic fields transform according to special relativity. The exact Lorentz transformations are:

$$\mathbf{E}' = (1-\gamma)\left(\frac{\mathbf{u}\cdot\mathbf{E}}{u^2}\right)\mathbf{u} + \gamma\left(\mathbf{E} + \mathbf{u}\times\mathbf{B}\right) \tag{74}$$

$$\mathbf{B}' = (1-\gamma)\left(\frac{\mathbf{u}\cdot\mathbf{B}}{u^2}\right)\mathbf{u} + \gamma\left(\mathbf{B} - \frac{1}{c^2}\mathbf{u}\times\mathbf{E}\right) \tag{75}$$

with $\gamma = \left(1 - \frac{u^2}{c^2}\right)^{-1/2}$.

In the non-relativistic limit $u \ll c$, these simplify to:

$$\mathbf{E}' \approx \mathbf{E} + \mathbf{u}\times\mathbf{B}, \quad \mathbf{B}' \approx \mathbf{B}$$

B Derivation of the Induction Equation

Starting from Ohm's law for a moving conductor:

$$\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v}\times\mathbf{B})$$

and Faraday's law:

$$\frac{\partial\mathbf{B}}{\partial t} = -\nabla\times\mathbf{E}$$

Ampère's law (neglecting the displacement current):

$$\nabla\times\mathbf{B} = \mu_0\mathbf{J}$$

Justification for neglecting the displacement current: the ratio of the displacement current to the magnetic term is

$$\frac{\varepsilon_0\partial\mathbf{E}/\partial t}{\nabla\times\mathbf{B}/\mu_0} \sim \frac{E}{B}\frac{U}{c^2} \sim \frac{U^2}{c^2} \ll 1$$

so the displacement current can be safely ignored in MHD.

Substituting Ohm's law into Faraday's law and using the definition of magnetic diffusivity $\eta = 1/(\mu_0\sigma)$, we obtain:

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$$

This is the final induction equation used in magnetohydrodynamics.

C Frozen-in Field Line Derivation

To rigorously demonstrate the frozen-in condition in ideal MHD, we start with the induction equation in the limit of infinite conductivity ($\eta = 0$):

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B})$$

We express this in the Lagrangian (co-moving) frame using the total derivative:

$$\frac{D\mathbf{B}}{Dt} = \frac{\partial \mathbf{B}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{B}$$

Using the vector identity:

$$\nabla \times (\mathbf{v} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{v} - (\mathbf{v} \cdot \nabla)\mathbf{B} - \mathbf{B}(\nabla \cdot \mathbf{v}) + \mathbf{v}(\nabla \cdot \mathbf{B})$$

and applying $\nabla \cdot \mathbf{B} = 0$, we substitute this into the induction equation and obtain:

$$\frac{D\mathbf{B}}{Dt} = (\mathbf{B} \cdot \nabla)\mathbf{v} - \mathbf{B}(\nabla \cdot \mathbf{v})$$

This is the Lagrangian evolution of the magnetic field. Meanwhile, from the continuity equation:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad \Rightarrow \quad \frac{D}{Dt}\left(\frac{1}{\rho}\right) = \frac{\nabla \cdot \mathbf{v}}{\rho}$$

Using the product rule:

$$\frac{D}{Dt}\left(\frac{\mathbf{B}}{\rho}\right) = \frac{1}{\rho}\frac{D\mathbf{B}}{Dt} - \frac{\mathbf{B}}{\rho^2}\frac{D\rho}{Dt}$$

Substituting the expressions for $\frac{D\mathbf{B}}{Dt}$ and $\frac{D\rho}{Dt}$, we arrive at:

$$\frac{D}{Dt}\left(\frac{\mathbf{B}}{\rho}\right) = \left(\frac{\mathbf{B}}{\rho}\right) \cdot \nabla \mathbf{v}$$

This evolution equation has the same form as that for a material line element $\delta\mathbf{r}$ in the fluid:

$$\frac{D\delta\mathbf{r}}{Dt} = \delta\mathbf{r} \cdot \nabla\mathbf{v}$$

Thus, $\mathbf{B}/\rho$ evolves like $\delta\mathbf{r}$. This implies that magnetic field lines deform and stretch in the same manner as the fluid elements they are embedded in illustrating the frozen-in condition. This result is the mathematical statement of **Alfvén's theorem**, which asserts that magnetic flux through a comoving surface is conserved in an ideal, perfectly conducting plasma.

D Derivation of Cooling Rates

D.1 Bremsstrahlung Cooling

Bremsstrahlung cooling arises from the deceleration of charged particles, particularly electrons, in the electric field of ions. The emissivity for thermal bremsstrahlung Shapiro and Teukolsky [2008]) is given by:

$$\Lambda_{ff} = 1.43 \times 10^{-27} \left(\frac{\rho}{m_p}\right)^2 T_e^{1/2} \quad \mathrm{erg\,cm^{-3}\,s^{-1}} \tag{76}$$

where, T_e is the temperature of electrons, m_e and m_p are the masses of electrons and protons.

The above equation is not normalized according to the dimensionless convention we have adopted, so we must turn it into the dimensionless form as follows.

To begin with, we must multiply the above equation by h, since the above equation gives the emissivity per unit volume and we want to determine the vertically integrated cooling rate. We also know the temperature of the proton is given by,

$$T_p = \frac{\mu m_p}{k_b \gamma} \frac{\beta}{\beta + 1}, \tag{77}$$

which is related to the temperature of electrons by relation from Chattopadhyay and Chakrabarti [2002], as:

$$T_e = \sqrt{m_e/m_p} T_p.$$

So using both, we may write the cooling rate as:

$$Q_{Br}^{-} = 1.43 \times 10^{-27}(\rho.\rho)\left(\frac{1}{m_p}\right)^2 ha\left(\frac{\mu m_p}{k_b\gamma}\right)^{1/2}\left(\frac{\beta}{\beta+1}\right)^{1/2} \quad \mathrm{erg\,cm^{-2}\,s^{-1}}, \tag{78}$$

also using 17 and 5, we can write mass accretion rate as,

$$\dot{M} = 4\pi I_n \rho h x v, \tag{79}$$

which can be expressed in units of $\dot{M}_{edd}$, Eddington rate as $\dot{m}$, or we can write,

$$\dot{M} = \dot{m} \times M_{edd}, \tag{80}$$

where Eddington rate is given as,

$$\dot{M}_{edd} = 1.39 \times 10^{17}\,\frac{M_{BH}}{M_\odot}\,\mathrm{g\,s^{-1}}. \tag{81}$$

Using it and substituting ρ, we can write the cooling rate as,

$$Q_{Br}^{-} = \frac{1.43 \times 10^{-27}}{m_p^2}\frac{\rho h(\dot{m} \times \dot{M}_{edd})a}{4\pi I_n h x v}\left(\frac{m_p}{m_e}\right)^{1/4}\left(\frac{\mu m_p}{k_b\gamma}\right)^{1/2}\left(\frac{\beta}{\beta+1}\right)^{1/2}\mathrm{erg\,cm^{-2}\,s^{-1}}. \tag{82}$$

Substituting h from 15 and Eddington rate from 81, we get,

$$Q_{Br}^{-} = \frac{1.98 \times 10^{-10}}{m_p^2}\frac{\rho h\dot{m}}{4\pi I_n x^{3/2}(x-1)v}\left(\frac{m_p}{m_e}\right)^{1/4}\left(\frac{\mu m_p}{2k_b}\right)^{1/2}\left(\frac{\beta}{\beta+1}\right)^{1/2}\frac{M_{BH}}{M_\odot}\mathrm{erg\,cm^{-2}\,s^{-1}}. \tag{83}$$

Now to make it dimensionless, we need to divide the whole equation by $2GM_{BH}c$, which gives the required result for cooling as,

$$\boxed{Q_{Br}^{-} = \frac{1.98 \times 10^{-10}}{m_p^2}\frac{\rho h\dot{m}}{4\pi I_n x^{3/2}(x-1)v}\left(\frac{m_p}{m_e}\right)^{1/4}\left(\frac{\mu m_p}{2k_b}\right)^{1/2}\left(\frac{\beta}{\beta+1}\right)^{1/2}\frac{1}{2GM_\odot c}}. \tag{84}$$

D.2 Synchrotron Cooling

The emissivity for optically thin synchrotron radiation in a magnetized relativistic plasma is given by Shapiro and Teukolsky [2008]):

$$\Lambda_{syn} = \frac{16}{3}\frac{e^2}{c}\left(\frac{eB}{m_e c}\right)^2\left(\frac{k_b T_e}{m_e c^2}\right)^2 n_e. \tag{85}$$

and using pressure relationship,

$$\frac{\langle B_\phi^2\rangle}{8\pi} = p_m = \frac{P}{\beta+1} \tag{86}$$

Assuming $n_e = n_p = \rho/m_p$, and expressing B via the magnetic pressure relation 86, T_e from previous relations, and multiplying by h, we write:

$$Q^-_{syn} = \frac{16}{3} 8\pi m_p \frac{e^4}{m_e^3 c^7}\left(\frac{P}{\beta+1}\right)\left(\frac{\beta}{\beta+1}\right)^2\left(\frac{\mu}{\gamma}\right)^2 a^4 n_e h. \tag{87}$$

Substituting $P = \rho a^2/\gamma$, expressing ρ and h from mass accretion rate, and simplifying:

$$Q^-_{syn} = \frac{32}{3} m_p \frac{e^4}{m_e^3 c^7}\frac{\beta^2}{(\beta+1)^3}\mu^2 \frac{\dot{M} a^5 \rho h}{v I_n \gamma^{5/2}\sqrt{2} x^{3/2}(x-1)}. \tag{88}$$

Switching to dimensionless accretion rate $\dot{m}$, substituting $a = a \cdot c$, and dividing by $2GM_{BH}c$, we get:

$$\boxed{Q^-_{syn} = 1.048\times 10^{18}\frac{e^4}{m_e^3}\frac{\beta^2}{(\beta+1)^3}\frac{\mu^2 \dot{m} a^5 \rho h}{v I_n \gamma^{5/2} x^{3/2}(x-1)}\frac{1}{2GM_\odot c}}. \tag{89}$$

Here, a is the adiabatic sound speed, and γ is the Lorentz factor associated with the flow. All constants are in cgs units, and the entire expression is normalized using $GM_\odot$ and c to match other dynamical quantities.

These derived forms are applicable in the optically thin limit and are used in the radial energy balance equation to assess radiative losses in magnetized accretion flows.

E Vertical Hydrostatic Equilibrium and Vertically Integrated Quantities

To determine the vertical half-thickness $h(x)$ of the accretion disk, we consider vertical hydrostatic equilibrium where the vertical velocity $v_z = 0$. The z-component of the momentum equation reduces to:

$$\frac{1}{\rho}\frac{\partial P}{\partial z} = -\frac{\partial \Phi}{\partial z}. \tag{90}$$

Using the Paczyński-Wiita potential,

$$\Phi(x, z) = -\frac{1}{2\left(\sqrt{x^2+z^2}-1\right)},$$

and applying the thin disk approximation ($z \ll x$), we expand $\sqrt{x^2+z^2} \approx x\left(1+\frac{z^2}{2x^2}\right)$. Substituting this into the potential and simplifying, the vertical component of the gravitational force becomes approximately:

$$\frac{\partial \Phi}{\partial z} \approx \frac{z}{2x(x-1)^2}.$$

$$\frac{1}{\rho}\frac{\partial P}{\partial z} = -\frac{z}{2x(x-1)^2}.$$

Integrating and applying the boundary condition $\rho(z=h) = 0$, we obtain:

$$h = \sqrt{\frac{2}{\gamma}}\, a x^{1/2}(x-1), \tag{91}$$

where $a = \sqrt{\frac{\gamma P}{\rho}}$ is the adiabatic sound speed.

To vertically integrate the density and pressure, we assume a polytropic vertical profile:

$$\rho(z) = \rho_0\left(1-\frac{z^2}{h^2}\right)^n, \quad P(z) = P_0\left(1-\frac{z^2}{h^2}\right)^{n+1},$$

where $n = 1/(\gamma-1)$ is the polytropic index, and ρ_0, P_0 are midplane values.

The vertically integrated quantities are then:

$$\Sigma = \int_{-h}^{h} \rho(z)\, dz = 2I_n \rho h, \tag{92}$$

$$W = \int_{-h}^{h} P(z)\, dz = 2I_{n+1} P h, \tag{93}$$

where the integration coefficients I_n and I_{n+1} are:

$$I_n = \int_0^1 (1-\xi^2)^n \, d\xi = \frac{(2^n n!)^2}{(2n+1)!}, \quad \text{with } \xi = \frac{z}{h}.$$

These coefficients provide analytical expressions for the integrated surface density and pressure, crucial for simplifying the governing equations in vertically averaged disk models.

F Divergence of the Stress Tensor in Cylindrical Coordinates

The divergence of a second-rank tensor $\mathbf{M}$ in cylindrical coordinates (x, ϕ, z), written in terms of its physical components, is given by:

Here, $\mathbf{M}$ denotes the stress tensor with components M_{ij}, and the divergence $\nabla \cdot \mathbf{M}$ represents the force density (per unit volume) arising from stresses in the fluid. The individual components are:

$$(\nabla \cdot \mathbf{M})_x = \frac{1}{x}\frac{\partial}{\partial x}(x M_{xx}) - \frac{M_{\phi\phi}}{x} + \frac{\partial M_{xz}}{\partial z}, \tag{94}$$

$$(\nabla \cdot \mathbf{M})_\phi = \frac{1}{x^2}\frac{\partial}{\partial x}(x^2 M_{\phi x}) + \frac{\partial M_{\phi z}}{\partial z}, \tag{95}$$

$$(\nabla \cdot \mathbf{M})_z = \frac{1}{x}\frac{\partial}{\partial x}(x M_{zx}) + \frac{\partial M_{zz}}{\partial z}. \tag{96}$$

These expressions describe the spatial variation of stresses in the radial (x), azimuthal (ϕ), and vertical (z) directions. In accretion disk models, particularly under the thin-disk approximation, vertical gradients ($\partial/\partial z$) are often neglected due to the disk's small height compared to radial extent. Nevertheless, all three components are essential for a complete representation. These components are used in evaluating the viscous and magnetic stresses in the momentum conservation equations within the magnetohydrodynamic framework.

G Full Derivation of Shock Conditions

This appendix contains the complete derivation of the shock conditions used in Section 4.1, including all intermediate algebraic and physical steps. The framework assumes a steady, axisymmetric, vertically integrated, magnetized accretion flow with toroidal magnetic field topology.

G.0.1 Mass Flux

We know mass flux from equation 5

$$\dot{M} = 2\pi\Sigma x v, \tag{97}$$

which at shock location x_s gives,

$$\dot{M}_- = \dot{M}_+ \tag{98}$$
$$\Sigma_- v_- = \Sigma_+ v_+. \tag{99}$$

Lets take J as some conserved quantity,

$$J = \Sigma v, \tag{100}$$

where $\Sigma \mathrm{and} v$ are vertically integrated pressure and inflow speed.

G.0.2 Momentum flux

Conservation of momentum flux gives us,

$$W_- + \Sigma_- v_-^2 = W_+ + \Sigma_+ v_+^2, \tag{101}$$

Lets take some Π as another conserved quanti across shock given as,

$$\Pi = W + \Sigma v^2. \tag{102}$$

By using momentum and mass flux we can get a relation for sound speed as,

$$\frac{\Pi v}{J} = \frac{W}{\Sigma} + v^2 \tag{103}$$
$$\frac{W}{\Sigma} = \frac{\Pi v}{J} - v^2 \tag{104}$$
$$g\frac{a^2}{\gamma} = \frac{\Pi v}{J} - v^2. \tag{105}$$

G.0.3 Magnetic flux conservation

From equation 49 and 50 we get magnetic flux as,

$$\dot{\Phi} = -\sqrt{4\pi} v h B_0(x) \tag{106}$$

where $B_0(x) = 2^{5/4} \pi^{1/4} \left(\frac{RT}{\mu}\right)^{1/2} \Sigma^{1/2} h^{-1/2} \beta^{-1/2}$

where at shock location $\dot{\Phi}_- = \dot{\Phi}_+$.

Using these equations we get,

$$\dot{\Phi} = -\sqrt{4\pi} v h 2^{5/4} \pi^{1/4} \left(\frac{RT}{\mu}\right)^{1/2} \Sigma^{1/2} h^{-1/2} \beta^{-1/2} \tag{107}$$

$$= -\sqrt{4\pi} v h^{1/2} 2^{5/4} \pi^{1/4} \left(\frac{RT}{\mu}\right)^{1/2} \Sigma^{1/2} \beta^{-1/2}. \tag{108}$$

Substituting h from equation 15 and using the fact that $\frac{RT}{\mu} = p_\pi = P\beta/(\rho(1+\beta))$ we can write above equation as,

$$\dot{\Phi} = -\sqrt{4\pi} v \left(\sqrt{\frac{2}{\gamma}} a x^{1/2}(x-1)\right)^{1/2} 2^{5/4} \pi^{1/4} \left(\frac{P\beta}{(\rho(1+\beta))}\right)^{1/2} \Sigma^{1/2} \beta^{-1/2}. \tag{109}$$

Now using equation 16 and $J = \Sigma v$ we can write,

$$\dot{\Phi} = -\sqrt{4\pi} \left(\sqrt{\frac{2}{\gamma}} x^{1/2}(x-1)\right)^{1/2} 2^{5/4} \pi^{1/4} \frac{a^{3/2}}{\sqrt{\gamma}} \frac{1}{\sqrt{\beta+1}} J^{1/2} v^{1/2}. \tag{110}$$

Using the relation we derived for sound speed at shock location (i.e. equation 105) we can write,

$$\dot{\Phi} = -\sqrt{4\pi} \left(\sqrt{2} x^{1/2}(x-1)\right)^{1/2} 2^{5/4} \pi^{1/4} \left(\frac{\gamma}{g}\left(\frac{\Pi v}{J} - v^2\right)\right)^{3/4} \frac{1}{\gamma^{3/4}} \frac{1}{\sqrt{\beta+1}} J^{1/2} v^{1/2}. \tag{111}$$

Since at shock location $x = x_s$ we get,

$$\dot{\Phi} = -\sqrt{4\pi}\left(\sqrt{2}x_s^{1/2}(x_s-1)\right)^{1/2} 2^{5/4}\pi^{1/4}\left(\frac{\gamma}{g}\left(\frac{\Pi v}{J} - v^2\right)\right)^{3/4}\frac{1}{\gamma^{3/4}}\frac{1}{\sqrt{\beta+1}}J^{1/2}v^{1/2}. \tag{112}$$

If we take some K' and $K = K'^2$ as,

$$K' = \sqrt{\frac{Jv}{\beta+1}}\left(\frac{\Pi v}{J} - v^2\right)^{3/4}\frac{1}{g^{3/4}} \tag{113}$$

$$K = \frac{Jv}{\beta+1}\left(\frac{\Pi v}{J} - v^2\right)^{3/2}\frac{1}{g^{3/2}}, \tag{114}$$

then from this equation, we can get shock condition as,

$$K_- = K_+ \tag{115}$$

$$\beta + 1 = \frac{1}{K}Jv\left(\frac{\Pi v}{J} - v^2\right)^{3/2}\frac{1}{g^{3/2}} \tag{116}$$

$$\beta + 1 = \frac{1}{K}Jv\frac{a^3}{\gamma^{3/2}}. \tag{117}$$

G.0.4 Energy flux conservation

Energy flux can be obtained by integrating radial momentum equation 22 as follows,

$$\mathcal{E} = \frac{v^2}{2} + \frac{a^2}{\gamma-1} + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} + \frac{\langle B_\phi^2\rangle}{4\pi\rho}. \tag{118}$$

Using the relation:

$$\frac{\langle B_\phi^2\rangle}{8\pi} = p_m \tag{119}$$

$$= \frac{P}{\beta+1}$$

$$\frac{\langle B_\phi^2\rangle}{4\pi\rho} = \frac{2P}{\rho(\beta+1)} \tag{120}$$

$$= \frac{2a^2}{\gamma}\left(\frac{1}{\beta+1}\right). \tag{121}$$

Therefore we have,

$$\mathcal{E} = \frac{v^2}{2} + \frac{a^2}{\gamma - 1} + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} + \frac{2a^2}{\gamma}\left(\frac{1}{\beta+1}\right). \tag{122}$$

Now substituting $\beta + 1$ from equation 116,

$$\mathcal{E} = \frac{v^2}{2} + \frac{a^2}{\gamma - 1} + \frac{\lambda^2}{2x^2} - \frac{1}{2(x-1)} + \frac{2a^2}{\gamma}\left(\frac{K\gamma^{3/2}}{Jva^3}\right), \tag{123}$$

$$\mathcal{E} - \frac{\lambda^2}{2x^2} + \frac{1}{2(x-1)} = \frac{v^2}{2} + \frac{a^2}{\gamma - 1} + \frac{2a^2}{\gamma}\left(\frac{K\gamma^{3/2}}{Jva^3}\right) \tag{124}$$

$$\mathcal{E} - \frac{\lambda^2}{2x^2} + \frac{1}{2(x-1)} = \frac{v^2}{2} + \frac{a^2}{\gamma - 1} + \frac{2}{\gamma}\left(\frac{K\gamma^{3/2}}{Jva}\right), \tag{125}$$

let $\mathcal{E} - \frac{\lambda^2}{2x^2} + \frac{1}{2(x-1)} = \mathcal{E}'$ then we can write above equation as,

$$\mathcal{E}' = \frac{v^2}{2} + \frac{a^2}{\gamma - 1} + \left(\frac{K}{J}\frac{2}{va}\gamma^{1/2}\right) \tag{126}$$

$$\frac{K}{J}\frac{2\gamma^{1/2}}{va} = \mathcal{E}' - \frac{v^2}{2} - \frac{a^2}{\gamma - 1}. \tag{127}$$

substituting a in this equation will give us the equation for $\mathcal{E}'$ which is conserved across the shock,

$$\boxed{\mathcal{E}' = \frac{v^2}{2} + \frac{1}{\gamma - 1}\frac{\gamma}{g}\left(\frac{\Pi}{J}v - v^2\right) + \frac{K}{J}\frac{2}{v}\frac{1}{\sqrt{\frac{1}{g}\left(\frac{\Pi}{J}v - v^2\right)}}} \tag{128}$$

where the following are the conserved quantities across shock,

$$\frac{\Pi}{J} = \frac{1}{v_-}\left(\frac{g \cdot a_-^2}{\gamma} + v_-^2\right) \tag{129}$$

$$\frac{K}{J} = \frac{1}{\beta_- + 1}\left(\frac{\Pi v_-}{J} - v_-^2\right)^{3/2}. \tag{130}$$

Now the eq. 128 can be solved to find the value of velocity after the jump takes place and with it we can find all other flow variables using conserved quantities $\frac{\Pi}{J}$ and $\frac{K}{J}$.

This concludes the full derivation of all shock conditions employed in the main text.

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